

Infinite-Dimensional Hahn Spectral Theory

Part I: infinitesimal thickening and range defects
in Hahn–Hilbert spaces

Part II: exact synthesis and algebra-relative spectra
for row- and column-finite Hahn matrices

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Abstract

This report assembles two independent infinite-dimensional spectral theories over one Hahn scalar field $K = \mathbb{C}((t^\Gamma))$. They are printed together because they continue the same non-claim of the collection’s finite-dimensional spectral report, and they are printed as two parts because the word *spectrum* does not mean the same thing in them, the vector spaces are different, and their hypotheses on Γ point in opposite directions.

Part I fixes an ordinary complex Hilbert space H and a nonzero set-sized *divisible* ordered abelian group Γ , and studies $\mathcal{H}_\Gamma = H((t^\Gamma))$ with its coefficientwise-convolution inner product. An automatic-structure theorem identifies every everywhere-defined adjointable operator, with no assumed continuity or support condition, with a Hahn series of bounded operators on H . For an ordinary bounded normal T , the set of $\lambda \in K$ for which $T - \lambda I$ fails to be bijective on $\mathcal{H}_\Gamma$ is exactly $\sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathbb{C}}(T)' + \mathfrak{m}_K)$, where the prime denotes ordinary accumulation points and $\mathfrak{m}_K$ the infinitesimals. Accumulation points thicken into whole infinitesimal monads; isolated points do not thicken; no new eigenvalues appear. An injective ordinary leading operator retains its entire algebraic range defect under every positive-order Hahn perturbation, so a coercive positive self-adjoint operator can fail to be onto with a cokernel of dimension at least the continuum.

Part II takes Γ *arbitrary*, with no divisibility assumption, and studies the algebra $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$ of Hahn series of row- and column-finite matrices acting on $\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$. For $A = D + B$ with $D = \text{diag}(d_i)$ of pairwise distinct ordinary real entries and $B = B^*$ of globally well-ordered positive support, there is a unique near-identity normalized Hahn-unitary diagonalization, exact vectorwise spectral synthesis, a full commutant and spectral-projection classification, and a coherent resolvent at every noneigenvalue. The spectrum relative to $\mathcal{A}_I$ is exactly the labeled eigenvalue set, even when the real residues accumulate. Weighted walks control every coefficient, and a sharp order $2r + 2$ finite-section threshold holds for one-scale locally finite graphs.

Section 3 states the interface: what the two spectra are, why the accumulation statements of the two parts are not in conflict, and which hypothesis is doing the work in each.

Status. Two candidate original theorem packages, each with explicit proofs and its own documented literature comparison. Neither claims to solve a named published open problem. Priority is not certified for either. The general proofs have not been independently refereed or checked by a proof assistant. The two accompanying programs check finite algebraic identities, not the infinite-dimensional theorems. Classical Hilbert, Hahn–Neumann, Higman and surreal normal-form inputs are identified rather than presented as new.

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Shared front matter

1 Why this is one report, and why it is two parts

1.1 The non-claim that both parts continue

The collection already contains a substantial finite-dimensional spectral report: Hermitian diagonalization, singular values, conditioning, determinantal scales, spectral descent, and ramification of positive matrix roots [2]. Every dimension there is an ordinary finite integer. That report closes by listing, in its own words, what it does *not* contain:

“There is no theorem here about infinite-dimensional Hilbert spaces over $\mathbf{No}[i]$, bounded-operator spectra, compact or trace-class operators, or spectral measures.”

The two manuscripts assembled here supply the first two of those items and the compact half of the third. Trace-class operators and spectral measures are left alone.

- *Infinite-dimensional Hilbert spaces over the surcomplex numbers.* Sections 5 and 6 construct one explicit such space and determine its adjointable operator algebra exactly.
- *Bounded-operator spectra.* Theorem 8.2 computes the spectrum of the scalar extension of every ordinary bounded normal operator.
- *Compact operators.* Corollary 9.2 does the same for ordinary compact normal operators, and states explicitly that “compact” there describes the original operator on the ordinary Hilbert space and asserts nothing about compactness or compactoidness in the Hahn valuation topology.
- *Trace-class operators and spectral measures* remain untouched. Nothing below is a trace-class theory. Theorem 19.2 gives a complete Boolean algebra of spectral projections, but remark 19.4 shows that the family of rank-one projections is not strongly Hahn summable as a family of *operators*, so that calculus is not a spectral measure and is not offered as one.

The finite-dimensional report is already a long treatise on determinantal finite-matrix theory. Absorbing these two manuscripts into it would bury both. This report is its sibling, not its continuation, and section 3 records what the two reports may and may not be read as saying jointly.

1.2 The central hazard: two senses of “spectrum”

The two parts use the word *spectrum* for two different things, on two different vector spaces, under opposite hypotheses on Γ . Quoted side by side without their categories, their conclusions about accumulation read as a contradiction. They are not one. No attempt is made below to write a single hypothesis paragraph covering both parts, and the divisibility hypothesis is *not* unified.

Section 3 works the apparent conflict out on a single operator and says exactly why both conclusions hold.

1.3 Part I at a glance

Throughout Part I, $H \neq 0$ is a complex Hilbert space, Γ is nonzero, divisible, ordered, abelian and set-sized, and

$$K = \mathbb{C}((t^\Gamma)), \quad \mathcal{H}_\Gamma = H((t^\Gamma)).$$

	Part I	Part II
Vector space	$\mathcal{H}_\Gamma = H((t^\Gamma))$, H an ordinary complex Hilbert space; every ordinary square-summable vector is a constant of $\mathcal{H}_\Gamma$	$\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$; each Hahn coefficient has <i>finite</i> coordinate support, but a vector may have infinitely many nonzero Hahn coordinates
Operator class	$\mathcal{A}_\Gamma(H) = \mathcal{B}(H)((t^\Gamma))$, Hahn series of <i>ordinary bounded</i> operators on H	$\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$, Hahn series of row- and column-finite matrices, with no ordinary boundedness at all
“Spectrum” means	$\sigma_\Gamma^{\text{alg}}(A) = \{\lambda \in K : A - \lambda I \text{ is not bijective on } \mathcal{H}_\Gamma\}$; for $A \in \mathcal{A}_\Gamma(H)$ also $\sigma_\Gamma^{\text{adj}}(A)$, failure of invertibility in $\mathcal{A}_\Gamma(H)$	$\sigma_{\mathcal{A}_I}(X) = \{z \in K : X - z\mathbf{1} \text{ has no two-sided inverse inside } \mathcal{A}_I\}$: a spectrum relative to a named unital K -algebra
Hypothesis on Γ	nonzero, set-sized, ordered abelian and divisible as a standing convention; the norm itself exists without divisibility, and section 12 further restricts Γ to a subgroup of $\mathbb{R}$	nonzero, set-sized, ordered abelian, not assumed divisible ; the source manuscript records the absence of that hypothesis as a feature, and only near-one binomial roots are used
What accumulation does	accumulation points of $\sigma_\mathbb{C}(T)$ acquire whole complex infinitesimal monads (theorem 8.2); isolated points do not thicken	accumulation of the ordinary residues d_i does not enlarge $\sigma_{\mathcal{A}_I}(A)$ beyond $\{\lambda_i\}$ (theorem 20.2 and example 20.3)

Table 1: The two senses of “spectrum” used in this report. Every spectral statement below names its vector space and which of these definitions it uses.

The valuation is the least support exponent. A prime on an ordinary compact set denotes its accumulation set, not a derivative and not an essential spectrum.

Theorem 1.1 (Part I theorem package). *The following assertions hold in this model.*

- (i) *Every everywhere-defined adjointable K -linear operator on $\mathcal{H}_\Gamma$ has a unique expansion in $\mathcal{B}(H)((t^\Gamma))$. Its coefficient operators are ordinary bounded operators and have one common well-ordered support.*
- (ii) *For an ordinary bounded normal T , its coefficientwise extension has*

$$\sigma_\Gamma(T) = \sigma_\mathbb{C}(T) \cup (\sigma_\mathbb{C}(T)' + \mathfrak{m}_K), \quad \sigma_{p,\Gamma}(T) = \sigma_{p,\mathbb{C}}(T). \quad (1.1)$$

Here σ_Γ is the failure-of-bijectivity spectrum on $\mathcal{H}_\Gamma$; it also equals the spectrum in the adjointable algebra.

- (iii) *If $S : V \rightarrow V$ is injective and E is an endomorphism-valued Hahn series of strictly positive order, then*

$$\text{coker}(S + E) \cong W((t^\Gamma)), \quad V = \text{Ran } S \oplus W,$$

noncanonically. No commutation hypothesis is required.

- (iv) *For $De_n = n^{-1}e_n$ and $s = t^\eta$, $\eta > 0$, the operator $C = D^2 + s^2I$ on $\ell^2((t^\Gamma))$ is bounded, positive, self-adjoint and coercive, but not surjective. Its cokernel has K -dimension at least $2^{\aleph_0}$. In rank one its range is closed and proper, with zero orthogonal complement.*
- (v) *The extension of an ordinary bounded T has a least F -valued norm bound if and only if T attains its ordinary Hilbert operator norm. When the least bound exists, it is the ordinary norm itself.*

Each assertion is proved in Part I. The result about the least norm bound is not a claim that these operators are unbounded: all operators in (i) have field-valued bounds. The failure is the absence of a *least* such bound. Likewise, a spectral point in (ii) need not be an eigenvalue.

1.4 Part II at a glance

Throughout Part II, Γ is a set-sized ordered abelian group with no divisibility assumption, I is a nonempty index set, and

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)), \quad \mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma)), \quad \mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma)).$$

Let $D = \text{diag}(d_i)$ with the $d_i \in \mathbb{R}$ pairwise distinct, let $B = B^* \in \mathcal{A}_I$ have positive support, and set $A = D + B$.

Theorem 18.2 constructs a unique pair (U, Λ) with

$$AU = U\Lambda, \quad U^*U = UU^* = \mathbf{1}, \quad U = \mathbf{1} + \text{positive-support terms},$$

where $\Lambda = \text{diag}(\lambda_i)$, $\lambda_i - d_i$ has positive support, and every U_{ii} is positive real Hahn. All exponents of the corrections belong to the additive monoid generated by $\text{supp } B$; the construction uses no extension of the value group. The distinct-real-label hypothesis entails $|I| \leq |\mathbb{R}|$; no claim of a larger residue index set is implicit.

Theorems 19.1, 19.2 and 20.2 then give exact eigenvector expansions, classify every commuting operator and spectral projection, and prove $\sigma_{\mathcal{A}_I}(A) = \{\lambda_i : i \in I\}$, the spectrum relative to the specified unital K -algebra and not a Banach-algebra spectrum. Theorems 21.2 to 21.4 and 21.6 determine how much of an infinite operator a single coefficient can see: an eigenvalue coefficient is supported by closed weighted walks, a change outside a finite section can first affect it only through a walk that leaves the section and returns, and the corresponding valuation threshold is exact in a family of path examples.

1.5 What is and is not claimed as new

Neither part claims a classical input as new, and the two parts have separate novelty boundaries and separate literature comparisons.

For Part I: the ordinary spectral theorem, Baire category, the closed graph theorem, Hilbert norm estimates and Hahn support lemmas are classical inputs [7, 5]. The non-Archimedean failure of orthogonal decomposition is known in other models, including work of Aguayo and Nova [8] and Aguayo, Nova and Shamseddine [9]; we do not claim to have discovered that failure in general. The proposed contribution is the particular combination of automatic adjoint structure, the exact accumulation-monad spectrum, the arbitrary positive-support range-defect formula, and the explicit coercive example with a continuum-dimensional defect in $H((t^\Gamma))$. In the repository sections and primary literature inspected, these statements were not found in this form. This is a qualified report of the search, not a proof of priority; a theorem may have predecessors under terminology not located by that search.

For Part II: simple-eigenvalue perturbation theory and its normalization conventions are classical, and Greenbaum, Li and Overton give an accessible account of both eigenvalue and eigenvector perturbation [13]; fixed-point and near-diagonal formulations are also established, including the work of Kenmoe, Smerlak and Zadorin [14]. The recursive Rayleigh–Schrödinger mechanism and the usual first corrections are not claimed as new. The support argument uses Neumann’s ordered-series machinery [5] and Higman’s word-order theorem [6]. Matrix products and walks have a long-established connection; the path-sum framework of Giscard, Thwaite and Jaksch is an explicit reference [15], and the walk theorem below is not a claim to have discovered that connection. The coefficient algebra is likewise not new: Ara, Goodearl and

O’Meara study real and complex row- and column-finite matrices and emphasize that positivity in that algebra does not force the existence of eigenvalues or square roots [16], which is why an infinite-dimensional result must state both its vector space and its operator algebra. The elementary no-eigenvector example in section 24 is a boundary test, not a new phenomenon. The proposed contribution is the combined arbitrary-rank Hahn theorem: a globally coherent normalized diagonalizer for infinitely many simple residue levels, exact synthesis on a specified Hahn module, coherent resolvents at every noneigenvalue, and marked-walk stability with sharp finite-section thresholds. Neither the finite report inspected in [3] nor the cited perturbation and row-finite sources state that package. This is a targeted comparison, not an exhaustive theorem of bibliographic absence; some components may have predecessors in formal operator or valued-algebra literature not located in this search.

Strand	Established input or predecessor	Contribution here
Finite surcomplex matrices	Repository spectral chapter	Infinite coefficient Hilbert spaces (Part I); arbitrary index sets (Part II)
Hahn formal calculus	Neumann support lemma	Automatic global support for adjoints (Part I)
Ordinary normal spectra	Hilbert spectral theorem	Exact infinitesimal thickening formula (Part I)
Formal perturbation	Neumann inversion near the identity	Persistence of arbitrary injective range defects (Part I)
Non-Archimedean orthogonality	Known nonorthomodular examples	A closed coercive range with continuum defect (Part I)
Operator norm bounds	Ordinary norm and attainment	Exact least-bound criterion over the Hahn field (Part I)
Perturbation recursions	Rayleigh–Schrödinger corrections	Global support certificate for all orders at once (Part II)
Row- and column-finite matrices	Ara–Goodearl–O’Meara	Hahn diagonalization under simple ordinary residues (Part II)
Matrix walks	Path-sum framework	Marked-walk stability and sharp section thresholds (Part II)

Table 2: The novelty boundary: classical tools are not renamed as new theorems.

1.6 Relation to the nearby non-Archimedean literature

Aguiayo–Nova–Shamseddine work with free Banach spaces of countable type over the complex Levi–Civita field, represented by weighted c_0 -type spaces [9]. Ishizuka proves spectral results for self-adjoint compactoid operators over valued fields with formally real residue field [10]. These are nearby but different settings. The constant copy of H in the Part I space retains every ordinary square-summable vector, while its valuation topology is discrete; the extension of an ordinary compact operator is not thereby compactoid in the new topology.

The June 2026 preprint of Miabey treats diagonal operators and finite-rank perturbations in a non-Archimedean Hilbert-space setting [12]. It is used only to delimit related contemporary work; none of its theorems is an input to the proofs here. Formal resolvent expansions also have a classical relationship with pole and Drazin-spectrum questions; see Boasso [11]. The normal-operator formula of Part I is proved directly, so no Drazin-theoretic equivalence is needed as an unproved bridge.

For Part II, the relevant nearby object is the algebra of ordinary row- and column-finite matrices itself [16]. The reference concerns ordinary such matrices and is not a source for the Hahn diagonalization asserted here.

1.7 Set-sized mathematics and terminology

All index sets, exponent groups, supports and families in the proofs are sets, and we work in ordinary ZFC. The passages to $\mathbf{No}[i]$ in sections 2.4 and 25 are transports of set-sized Hahn objects, not formations of a set containing all surreal numbers. The full surreal class offers additional scales but is not a set-sized coefficient field in ZFC; the proofs below avoid class-indexed summations and class-sized choices by fixing the workspace at the outset.

“Positive support” means that every exponent is strictly greater than zero; it does not mean that the complex coefficients are positive. “Hermitian” means $A^* = A$. In Part II, “unitary” means two-sided algebraic unitarity in $\mathcal{A}_I$, and a “Hahn-orthonormal basis” means a family with the strong-summation synthesis property proved in theorem 19.1, not a Hamel basis and not an ordinary Hilbert-space basis.

2 Shared Hahn preliminaries

This section fixes the conventions used by both parts and states the support lemma once. Everything else about the two constructions — their inner products, their operator algebras, their divisibility hypotheses — is stated separately in each part, because in each case the two differ.

2.1 Support conventions and strong summability

A Hahn series with coefficients in a complex vector space V is a family

$$x = \sum_{\gamma \in \Gamma} x_\gamma t^\gamma, \quad x_\gamma \in V,$$

whose support $\{\gamma : x_\gamma \neq 0\}$ is well ordered in the given order of Γ . Write $V((t^\Gamma))$ for the resulting set. The zero series has empty support. For $x \neq 0$ put $v(x) = \min \text{supp } x$, and put $v(0) = +\infty$.

A family $(x_a)_{a \in J}$ in $V((t^\Gamma))$ is *Hahn summable*, also called *strongly Hahn summable*, when its union of supports is well ordered and, for every γ , only finitely many a have $(x_a)_\gamma \neq 0$. Its sum, written $\sum_a^H x_a$, is defined coefficientwise by those finite sums. All unqualified infinite formal sums in this report mean this notion. It is not convergence of ordinary partial sums in the full surreal fine topology, and proposition 23.1 exhibits a strongly summable family whose partial sums do not converge even in the intrinsic valuation topology of K .

2.2 The support lemma, stated once

Both source manuscripts open with the Hahn–Neumann support lemma, one citing it and supplying the two-set finiteness argument, the other reproving Higman’s word-order input by a minimal bad sequence. The lemma is classical, and it is already printed with a full proof elsewhere in this collection: the finite-dimensional spectral report proves the word-order statement as its `spec:lem:words` and the positive-monoid statement as its `spec:lem:neumann`, in its section “A self-contained support engine” [2]. We therefore state it here once, in the form both parts use, and do not reprint that proof.

Lemma 2.1 (Classical Hahn–Neumann support lemma). *Let Γ be an ordered abelian group.*

- (i) *If $A, B \subseteq \Gamma$ are well ordered, then $A + B$ is well ordered and each γ has only finitely many representations $\gamma = a + b$ with $a \in A$ and $b \in B$. A finite union of well-ordered subsets is well ordered, and the corresponding statement holds for any finite number of summands.*

(ii) If $S \subseteq \Gamma_{>0}$ is well ordered, then the additive monoid

$$S^* = \{s_1 + \cdots + s_n : n \geq 0, s_j \in S\},$$

including the empty sum 0, is well ordered, and for each $\gamma \in \Gamma$ there are only finitely many ordered words $(s_1, \dots, s_n)$ over S with sum γ , the length n being allowed to vary.

This is the standard support lemma [5], whose word form rests on Higman's theorem that finite words over a well-quasi-ordered alphabet are well-quasi-ordered [6]. Part (i) is elementary and both source manuscripts give the argument: infinitely many distinct a in a fixed-sum family yield an increasing subsequence in A and hence a strictly decreasing subsequence $\gamma - a$ in B , a contradiction; and every sequence in a well-ordered set has an infinite nondecreasing subsequence, obtained by successively choosing an occurrence of the minimum of the remaining tail. Part (ii) is the usual Neumann lemma in its positive-support form. Both are stated classical inputs here, not new combinatorial claims.

Remark 2.2 (Why the full word form is needed). Both parts use the finite-word assertion of lemma 2.1(ii), not only the assertion that each fixed power is meaningful. The latter alone would not justify a geometric series. The estimate $v(X^n) \geq n\varepsilon$ is also not a substitute: in a value group of higher rank the sequence $n\varepsilon$ may be bounded above, as section 23 exhibits concretely.

Lemma 2.1(i) licenses scalar multiplication on $V((t^\Gamma))$ and products of operator-valued Hahn series, even when the coefficient algebra is noncommutative; a finite number of supports is combined by iteration.

2.3 Support-certified inversion and formal substitution

Lemma 2.3 (Support-certified Neumann inverse). *Let $\mathcal{R}$ be a unital complex algebra and let $E \in \mathcal{R}((t^\Gamma))$ have $v(E) > 0$. Then*

$$(I + E)^{-1} = \sum_{n \geq 0} (-E)^n \tag{2.1}$$

is a well-defined two-sided inverse in $\mathcal{R}((t^\Gamma))$, with support in $(\text{supp } E)^$.*

Proof. Expand each coefficient of E^n as a sum over ordered words in $\text{supp } E$. Lemma 2.1 makes the union well ordered and all coefficients finite, including the sum over n ; the products of the resulting series are the formal convolution coefficients, and the geometric identity is a formal one-variable identity. Noncommutativity causes no problem, because every factor is a power of the same E . $\square$

Remark 2.4 (A second proof, from the first source). The Part I manuscript proves lemma 2.3 by a different route, which is recorded here because it uses no formal-identity bookkeeping. After lemma 2.1 has made every coefficient of $\sum_{n \geq 0} (-E)^n$ a finite sum, the finite identity $(I + E) \sum_{n=0}^N (-E)^n = I + (-1)^N E^{N+1}$ telescopes coefficientwise across all word lengths, since at a fixed exponent only finitely many word lengths occur; right multiplication telescopes as well. This argument uses associativity but not commutativity of $\mathcal{R}$.

Corollary 2.5 (Evaluation of homogeneous constructions). *Let $\mathcal{R}$ be a unital complex algebra, let $S \subseteq \Gamma_{>0}$ be well ordered, and let $X_n \in \mathcal{R}((t^\Gamma))$ satisfy $\text{supp } X_n \subseteq nS$ for every $n \geq 1$. Then $(X_n)_{n \geq 1}$ is strongly summable. Finite sums, products, coefficientwise linear maps preserving supports, and regroupings of such homogeneous constructions commute with their evaluation $\sum_{n \geq 1}^H X_n$. The same holds with $\mathcal{R}$ replaced by a module over it, or by K itself.*

Proof. The union of supports lies in S^* . At a fixed exponent, lemma 2.1(ii) bounds the possible lengths and the possible ordered exponent words. All coefficient operations within one word are finite, so the claimed interchanges are finite rearrangements coefficient by coefficient. $\square$

Corollary 2.6 (Formal power series at a positive-support argument). *If $X \in \mathcal{R}((t^\Gamma))$ has positive support, then any one-variable formal power series with complex coefficients can be evaluated at X , and formal identities between such series remain valid whenever their substitutions have zero constant term. In particular $(I + X)^{-1}$ is given by (2.1) and $(I + X)^{-1/2}$ is defined by its binomial series.*

Proof. Take $S = \text{supp } X$ in corollary 2.5. $\square$

2.4 The surcomplex realization

Suppose an ordered additive embedding $j : \Gamma \hookrightarrow \mathbf{No}$ is fixed; when Γ is itself a set-sized ordered subgroup of the surreal additive group, take j to be the inclusion. Conway normal forms give the maps

$$\sum_{\gamma} r_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} r_{\gamma} \omega^{-j(\gamma)}, \quad r_{\gamma} \in \mathbb{R}, \quad (2.2)$$

and their coefficientwise complexification. These are an ordered-field embedding $F \hookrightarrow \mathbf{No}$ and a field embedding $K \hookrightarrow \mathbf{No}[i]$, preserving conjugation and the prescribed strong sums. The classical normal-form and multiplication facts are standard surreal theory [4].

The sign in (2.2) matters. A well-ordered set of Hahn exponents becomes a reverse-well-ordered set of Conway exponents, exactly the normal-form condition, and positive Hahn exponents become infinitesimal monomials. All supports are sets, so the normal forms define actual surreal or surcomplex numbers, and the vector spaces below never become proper classes. In particular, $\Gamma = \mathbb{Q}$ and $t = \omega^{-1}$ suffice for all explicit counterexamples of Part I.

Warning 2.7 (Distinct completions and topologies). The Part I construction $H((t^\Gamma))$ is neither the algebraic tensor product $H \otimes_{\mathbb{C}} K$ nor a silently chosen sequence space $\ell^2(K)$. The Part II space $\mathbb{C}^{(I)}((t^\Gamma))$ is likewise not ordinary ℓ^2 , and its vectors have finite-support *coefficients* without having finitely many nonzero Hahn coordinates. Nor does the intrinsic valuation topology of either space equal the topology induced by embedding its scalar field into the full surreal fine topology, in which the set-sized scalar workspace is discrete [1]. All of these distinctions matter to the results.

3 The interface between the two parts

3.1 What is shared and what is not

The two parts share the scalar field K , the support conventions of section 2.1, the support lemma lemma 2.1, the inversion tools of section 2.3, and the surcomplex embedding (2.2). Nothing else is shared. In particular no theorem of Part I is used in the proof of any theorem of Part II, and no theorem of Part II is used in Part I. Each part has its own inner product, its own operator algebra, its own spectral definition, and its own hypothesis on Γ , and each proves its own positivity statement from scratch: proposition 5.1 constructs the positive square root $\|x\| = \sqrt{\langle x, x \rangle}$ from the even leading exponent of $\langle x, x \rangle$ and a near-one binomial root; this works for every ordered value group. Likewise, proposition 17.3 proves positivity without assuming that every positive scalar has a square root. Part II uses no norm, and corollary 18.3 explicitly allows nondivisible groups.

The stated standing hypotheses are retained separately below; no hypothesis should be carried across the boundary. In particular, divisibility remains a Part I convention, but is not necessary

for its norm construction. General positive scalars and squared norms have different root requirements: t has no square root in $\mathbb{R}((t^{\mathbb{Z}}))$, whereas $\|te\| = t$ for any ordinary unit vector e .

3.2 The accumulation statements are about different objects

Read without their categories, the two parts appear to disagree about accumulation. Part I proves that accumulation points of the ordinary spectrum thicken into infinitesimal monads. Part II proves that the algebra-relative spectrum is exactly the labeled eigenvalue set even when the real residues accumulate. One operator makes the difference visible.

Let $I = \mathbb{N}_{\geq 1}$ and consider the diagonal data $d_n = 1/n$, whose residues accumulate at 0.

- As an *ordinary bounded operator* on $H = \ell^2(\mathbb{N}_{\geq 1})$, the map $De_n = n^{-1}e_n$ is compact, normal and injective, and its ordinary spectrum is $\{1/n : n \geq 1\} \cup \{0\}$ with 0 as its only accumulation point. Corollary 9.2 then gives, for the failure-of-bijectivity spectrum on $\mathcal{H}_\Gamma = \ell^2((t^\Gamma))$,

$$\sigma_\Gamma(D) = \mathfrak{m}_K \cup \{1/n : n \geq 1\}.$$

The whole infinitesimal monad of 0 is in the spectrum. The witness is concrete: a putative solution of $Dy = f$ with $f = (n^{-1})_n$ would need leading coefficient $(1)_n$, which is not in ℓ^2 .

- As an *element of $\mathcal{A}_I$* with $I = \mathbb{N}_{\geq 1}$ and $B = 0$, the same matrix $\text{diag}(1/n)$ has $\text{diag}(n) \in \text{RCF}_I(\mathbb{C}) \subseteq \mathcal{A}_I$ as a two-sided inverse, because a diagonal matrix of unbounded complex entries is a perfectly good row- and column-finite matrix. Hence $0 \notin \sigma_{\mathcal{A}_I}(D)$, and example 20.3 records this.

Both statements are true. They are not statements about the same operator: one is about a bounded operator on a Hilbert space whose vectors are square-summable, the other about a matrix acting on finite-support coefficient vectors, and invertibility is being tested in two different algebras. The unbounded diagonal inverse is excluded in Part I precisely because theorem 6.1 forces the coefficients of an adjointable operator to be ordinary *bounded* operators; it is admitted in Part II precisely because row- and column-finiteness replaces boundedness there. Section 14 makes the same point from the finite-section side, and proposition 24.1 makes the converse point: the Part II diagonalizer has a first coefficient with unbounded entries, so that theorem does not secretly solve the ordinary bounded-operator problem.

3.3 Reading either part alone

Each part is self-contained given section 2. A reader interested in the Hilbert-space obstructions can read Part I and stop; a reader interested in exact infinite diagonalization can read Part II and stop. Section 28 describes the two verification suites together, and appendices A and B record the two separate repository and literature audits, which were carried out against two different pinned revisions.

Part I. Hahn–Hilbert spaces: infinitesimal spectral thickening and closed-range defects

Standing hypotheses for Part I. $H \neq 0$ is an ordinary complex Hilbert space and Γ is a nonzero set-sized *divisible* ordered abelian group. This convention guarantees square roots of all positive scalars in $F = \mathbb{R}((t^\Gamma))$; the norm construction below needs only roots of squared norms and works without divisibility. Section 12 temporarily strengthens the hypothesis to $\Gamma \subseteq \mathbb{R}$. “Spectrum” in this part means failure of bijectivity on $\mathcal{H}_\Gamma = H((t^\Gamma))$, or failure of invertibility in $\mathcal{A}_\Gamma(H)$, and theorem 8.2 proves that for constant normal operators the two agree.

4 Why infinite dimension is the next substantial question

This part asks what happens when an ordinary infinite-dimensional Hilbert space is retained as the coefficient space, rather than replacing a finite matrix by a larger finite matrix. Over $\mathbb{C}$, positivity, completeness and adjoints support a familiar chain of implications: self-adjoint spectra are real, coercive operators are invertible, and closed subspaces admit orthogonal projections. The construction below retains a positive-definite inner product, all ordinary Hilbert vectors as constants, and a large adjointable operator algebra; in rank one it is complete. Nevertheless several of those implications fail, for an exactly identifiable reason: *a leading coefficient must itself be an ordinary Hilbert vector*.

The central results are not a list of isolated anomalies. An automatic operator-structure theorem makes the admissible algebra intrinsic. An injective-leading-term theorem explains the persistence of range defects. Together with ordinary normal spectral theory, these give an exact spectral classification and quantitative counterexamples to coercive inversion.

5 The Hahn–Hilbert space

5.1 The field and its divisibility hypothesis

Define

$$F = \mathbb{R}((t^\Gamma)), \quad K = \mathbb{C}((t^\Gamma)) = F[i].$$

Order F by its first nonzero coefficient and conjugate K coefficientwise. Let $\mathcal{O}_K = \{z : v(z) \geq 0\}$ and $\mathfrak{m}_K = \{z : v(z) > 0\}$, including zero in both sets. Coefficient extraction at zero gives $\text{st} : \mathcal{O}_K \rightarrow \mathbb{C}$. Every finite scalar has a unique expression $z = a + \varepsilon$ with $a \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$.

Divisibility of Γ guarantees positive square roots in F . For completeness, if $r = ct^\delta(1+u) > 0$, where $c > 0$ and $v(u) > 0$, then

$$\sqrt{r} = \sqrt{c}t^{\delta/2} \sum_{n \geq 0} \binom{1/2}{n} u^n.$$

The support lemma and the formal binomial identity prove its square is r ; the positive root is unique in any ordered field. Divisibility supplies $\delta/2$ for arbitrary positive scalars. For the norm, however, the leading exponent is already even, so no group extension is needed. The next proposition proves this directly. We retain the Part I standing hypotheses for the subsequent results; they are not a claim that divisibility is necessary to define the norm.

5.2 The Hahn–Hilbert inner product

We take ordinary and extended inner products to be linear in the first variable and conjugate-linear in the second. On $\mathcal{H}_\Gamma = H((t^\Gamma))$ define

$$\langle x, y \rangle = \sum_{\alpha, \beta} \langle x_\alpha, y_\beta \rangle_H t^{\alpha+\beta}. \quad (5.1)$$

Each ordinary Hilbert inner product in the coefficient is evaluated *before* the Hahn convolution. For example, a constant vector $(1/n)_{n \geq 1} \in \ell^2$ has its ordinary finite real squared norm, even though the separate family of its constant-coordinate squared magnitudes is not Hahn summable. We are not using that inadmissible family as a Hahn sum.

Proposition 5.1 (Positive-definite scalar extension). *For any ordered abelian group Γ , formula (5.1) defines a positive-definite Hermitian form. Its norm $\|x\| = \sqrt{\langle x, x \rangle}$ exists in F , satisfies the Cauchy–Schwarz and triangle inequalities, and*

$$v(\langle x, x \rangle) = 2v(x), \quad v(\|x\|) = v(x) \quad (x \neq 0). \quad (5.2)$$

If $x = t^\delta(u + \text{positive-order terms})$ with $u \neq 0$, then $\|x\| = t^\delta(\|u\|_H + \text{positive-order terms})$.

Proof. Lemma 2.1(i) gives well-definedness and sesquilinearity. The coefficient at 2δ , where $\delta = v(x)$, is exactly $\|x_\delta\|_H^2 > 0$; all other contributions have larger exponent. This proves positivity. For $x \neq 0$ write

$$\langle x, x \rangle = \|x_\delta\|_H^2 t^{2\delta} (1 + u), \quad u = 0 \text{ or } v(u) > 0.$$

The binomial root of corollary 2.6 gives

$$\|x\| = \|x_\delta\|_H t^\delta \sum_{n \geq 0} \binom{1/2}{n} u^n.$$

This is a positive square root in F for every Γ ; set $\|0\| = 0$. Its leading term proves both valuation assertions. The same argument for $H = \mathbb{C}$ defines the scalar modulus $|z| = \sqrt{z\bar{z}}$ without divisibility. Completing a square for $y \neq 0$ (the case $y = 0$ is immediate), using $\langle x - ay, x - ay \rangle \geq 0$ with $a = \langle x, y \rangle / \langle y, y \rangle$, gives $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$. The usual expansion of $\|x + y\|^2$, followed by this inequality, gives the triangle inequality. These are finite algebraic arguments over the ordered field F . $\square$

6 Every adjoint has a global Hahn expansion

6.1 The apparently smaller coefficient algebra

Let $\mathcal{A}_\Gamma(H) = \mathcal{B}(H)((t^\Gamma))$. Thus an element $A = \sum_\gamma A_\gamma t^\gamma$ has bounded ordinary operator coefficients and one well-ordered support. There is no uniform ordinary bound on the norms of all its coefficients. It acts by

$$(Ax)_\delta = \sum_{\gamma+\alpha=\delta} A_\gamma x_\alpha. \quad (6.1)$$

Lemma 2.1 proves this action is defined on every $x \in \mathcal{H}_\Gamma$. The series $A^* = \sum_\gamma A_\gamma^* t^\gamma$ satisfies $\langle Ax, y \rangle = \langle x, A^*y \rangle$, by finite coefficientwise rearrangement.

One might suspect that $\mathcal{A}_\Gamma(H)$ is only a convenient restricted operator class. The next theorem says that the existence of an everywhere-defined adjoint forces exactly this class, without any advance assumption that an operator preserves Hahn sums.

Theorem 6.1 (Automatic adjoint structure). *Let $A, B : \mathcal{H}_\Gamma \rightarrow \mathcal{H}_\Gamma$ be everywhere-defined K -linear maps such that*

$$\langle Ax, y \rangle = \langle x, By \rangle \quad (x, y \in \mathcal{H}_\Gamma). \quad (6.2)$$

There are unique bounded operators $A_\gamma \in \mathcal{B}(H)$ with well-ordered nonzero support such that A acts by (6.1). Moreover $B_\gamma = A_\gamma^$. Consequently*

$$\text{End}_{\text{adj}}(\mathcal{H}_\Gamma) \cong \mathcal{B}(H)((t^\Gamma))$$

as $$ -algebras. No separability of H is assumed.*

Proof. Step 1: bounded coefficient operators. For constant vectors $u, w \in H$, define $A_\gamma u = [t^\gamma]Au$ and $B_\gamma w = [t^\gamma]Bw$. Extracting the coefficient at γ in (6.2) gives

$$\langle A_\gamma u, w \rangle_H = \langle u, B_\gamma w \rangle_H. \quad (6.3)$$

Both maps are defined on all of H . To see that A_γ has closed graph, let $u_n \rightarrow u$ and $A_\gamma u_n \rightarrow z$ in the ordinary Hilbert norm. Testing against any fixed w in (6.3) gives $\langle z, w \rangle = \langle u, B_\gamma w \rangle = \langle A_\gamma u, w \rangle$. Hence $z = A_\gamma u$. The ordinary closed graph theorem makes A_γ bounded. The same argument applies to B_γ , and (6.3) identifies the two as adjoints.

Step 2: one well-ordered support. Let $S = \{\gamma : A_\gamma \neq 0\}$. If S were not well ordered, ordinary choice permits a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$ in S . Each $\ker A_{\gamma_n}$ is a proper closed subspace of H , and hence nowhere dense. Baire's theorem gives $u \in H$ outside their countable union. Then the support of Au contains every γ_n , contradicting its being a Hahn series. Thus S is well ordered. Because an ordinary bounded operator is zero exactly when its adjoint is zero, the coefficient family of B has the same support.

Step 3: constants determine the full operator. Let $\tilde{A}, \tilde{B}$ be the convolution operators built from these coefficient families. They are adjoints by lemma 2.1 and agree with A, B on constants. For arbitrary $x \in \mathcal{H}_\Gamma$ and constant $w \in H$,

$$\langle Ax, w \rangle = \langle x, Bw \rangle = \langle x, \tilde{B}w \rangle = \langle \tilde{A}x, w \rangle.$$

If a Hahn vector pairs to zero with every constant w , then each of its ordinary coefficients pairs to zero with all of H , so every coefficient vanishes. Hence $Ax = \tilde{A}x$. The analogous conclusion holds for B . Uniqueness follows by evaluating on constant vectors and extracting coefficients. Composition and the involution are the finite Hahn convolution operations, proving the algebra assertion. $\square$

Corollary 6.2 (Automatic summability and boundedness). *Every everywhere-defined adjointable operator preserves Hahn-summable families. It also admits a bound*

$$\|Ax\| \leq M\|x\| \quad (x \in \mathcal{H}_\Gamma) \quad (6.4)$$

for some $M \in F_{\geq 0}$. If $A \neq 0$ and $\gamma_0 = \min \text{supp } A$, any ordinary real $c > \|A_{\gamma_0}\|_H$ gives the valid bound $M = ct^{\gamma_0}$.

Proof. For a Hahn-summable family, the finiteness in lemma 2.1(i), applied with $\text{supp } A$, allows the two sums to be interchanged, and their union of supports remains well ordered. For the bound, write the leading term of x as $t^\delta u$. At exponent $2(\gamma_0 + \delta)$, the coefficient of $M^2\|x\|^2 - \|Ax\|^2$ is $c^2\|u\|_H^2 - \|A_{\gamma_0}u\|_H^2 > 0$. This remains true when $A_{\gamma_0}u = 0$, since then the leading output moves to a higher exponent. Thus the difference is positive for nonzero x . $\square$

Remark 6.3. The Baire argument is a uniform-support theorem, not a uniform bound on $\|A_\gamma\|_H$. The proof also explains why unrestricted ordinary algebraic endomorphisms, used later for choosing range complements, are not automatically adjointable.

7 An injective leading term preserves the whole range defect

7.1 A theorem independent of Hilbert structure

The next result applies to any complex vector space V . Write $V_\Gamma = V((t^\Gamma))$. An ordinary linear map on V extends coefficientwise, even if V also carries a topology and the map is discontinuous in that topology.

Theorem 7.1 (Positive-order defect rigidity). *Let $S : V \rightarrow V$ be injective. Let $E \in \text{End}_\mathbb{C}(V)((t^\Gamma))$ have strictly positive order, or be zero, and put $A = S + E$. Choose an algebraic complement*

$$V = \text{Ran } S \oplus W. \quad (7.1)$$

Then

$$V_\Gamma = A(V_\Gamma) \oplus W((t^\Gamma)). \quad (7.2)$$

In particular, A is injective, $v(Ax) = v(x)$ for $x \neq 0$, and $\text{coker } A \cong W((t^\Gamma))$ as K -vector spaces. The isomorphism depends on the chosen complement, but the nonvanishing of the defect does not.

Proof. At the least exponent δ of a nonzero x , the coefficient of Ax is $Sx_\delta \neq 0$. All contributions from E have greater exponent. This proves injectivity and the valuation assertion.

Define $U : V \rightarrow V$ by $U(Sv + w) = v$ for $v \in V$, $w \in W$. Then $US = I$ and $\ker U = W$. Extend U coefficientwise and use lemma 2.3 to set

$$R = (I + UE)^{-1}, \quad L = RU, \quad \Pi = I - AL.$$

The coefficients of U need not be bounded in any additional topology; all these formulas are legitimate in the ordinary algebraic endomorphism Hahn algebra. The identities

$$LA = R(US + UE) = R(I + UE) = I$$

and

$$U\Pi = U - UARU = U - (I + UE)RU = 0$$

show that Π kills the range of A and takes values in $\ker U = W((t^\Gamma))$. On that latter space L vanishes, so Π is the identity. Finally, every x decomposes as $ALx + \Pi x$; the intersection of the two summands is zero because $LA = I$ and $L|_{W((t^\Gamma))} = 0$. $\square$

Corollary 7.2 (No positive-order repair). *An injective nonsurjective ordinary operator cannot become surjective on V_Γ by adding any globally supported positive-order Hahn perturbation. This includes perturbations that do not commute with it and perturbations with infinitely many positive exponents.*

Corollary 7.3 (Explicit independent obstruction classes). *Suppose $(f_j)_{j \in J} \subset V$ have linearly independent classes in $V/\text{Ran } S$ over $\mathbb{C}$. Their constant Hahn extensions have linearly independent classes in $V_\Gamma/(S + E)V_\Gamma$ over K . Consequently*

$$\dim_K \text{coker}(S + E) \geq \dim_\mathbb{C}(V/\text{Ran } S).$$

Proof. In a nonzero finite sum $f = \sum_j c_j f_j$ with $c_j \in K$, take the least exponent among the nonzero c_j . The coefficient at that exponent is a nonzero finite complex combination of the f_j , and is not in $\text{Ran } S$. But the leading coefficient of $(S + E)x$ is always in $\text{Ran } S$. $\square$

Warning 7.4. Theorem 7.1 gives the actual vector-space quotient, not only a first-order approximation to it. It does *not* assert $\dim_K W((t^\Gamma)) = \dim_\mathbb{C} W$ for arbitrary infinite-dimensional W . The safe dimension statement is the lower bound just proved in corollary 7.3.

7.2 Why the result is more informative than a determinant test

There is no finite determinant for an arbitrary infinite-dimensional S . An entrywise infinitesimal perturbation may even make every finite section invertible, while the leading ordinary coefficient of a proposed inverse vector is excluded from H . The exact splitting (7.2) shows that this is not an accidental choice of data: the full algebraic cokernel persists. On the other hand, if S is an ordinary bijection, $S + E = S(I + S^{-1}E)$ is invertible by the same support-certified Neumann formula. Thus the injective case has a sharp dichotomy, not merely a negative example.

8 The exact spectrum of an ordinary normal operator

8.1 Two spectral definitions and one ordinary reduction

For a K -linear endomorphism A of $\mathcal{H}_\Gamma$, define

$$\sigma_\Gamma^{\text{alg}}(A) = \{\lambda \in K : A - \lambda I \text{ is not bijective on } \mathcal{H}_\Gamma\}.$$

For $A \in \mathcal{A}_\Gamma(H)$, define $\sigma_\Gamma^{\text{adj}}(A)$ by failure of invertibility in the unital algebra $\mathcal{A}_\Gamma(H)$. These definitions need not be identified in advance. In the theorem below, explicit inverses and explicit range obstructions will prove their equality. Neither is the algebra-relative spectrum $\sigma_{\mathcal{A}_I}$ of Part II, which is taken in a different algebra acting on a different vector space.

An ordinary bounded operator T is regarded as a constant member of $\mathcal{A}_\Gamma(H)$. We often suppress the extension subscript on T .

Lemma 8.1 (Normal isolated-point decomposition). *Let $T \in \mathcal{B}(H)$ be normal and $a \in \sigma_\mathbb{C}(T)$. Put $N = \ker(T - aI)$, $M = N^\perp$, and $S = (T - aI)|_M$. Then M, N reduce T and S is injective. If a is isolated in $\sigma_\mathbb{C}(T)$, then $N \neq 0$ and S is boundedly invertible on M . If a is an accumulation point, then S is not surjective.*

Proof. These facts follow from the ordinary normal spectral theorem [7]; here is the precise use of it. The spectral projection of $\{a\}$ is the orthogonal projection onto N . For isolated a , the function $(z - a)^{-1}$ on the rest of the compact spectrum is bounded, so spectral functional calculus gives an inverse on M . The projection cannot be zero, since otherwise a would not belong to the spectrum. For accumulation a , if S were onto, its injectivity and the bounded inverse theorem would give a bounded inverse. The spectrum of T would then be contained in $\{a\}$ together with a compact set separated from a , contradicting accumulation. This also covers $N = 0$. $\square$

8.2 Classification and all inverse formulas

Theorem 8.2 (Accumulation-monad spectrum). *For every ordinary bounded normal $T \in \mathcal{B}(H)$,*

$$\sigma_\Gamma^{\text{alg}}(T) = \sigma_\Gamma^{\text{adj}}(T) = \sigma_\mathbb{C}(T) \cup (\sigma_\mathbb{C}(T)' + \mathfrak{m}_K). \quad (8.1)$$

No scalar of negative valuation is in this spectrum. A nonzero infinitesimal displacement of an isolated ordinary spectral point is in the resolvent, whereas every infinitesimal displacement of an accumulation point remains in the spectrum.

Proof. We partition all scalars $\lambda \in K$ into exhaustive cases.

Infinite scalar. If $v(\lambda) < 0$, then

$$(T - \lambda I)^{-1} = - \sum_{n \geq 0} \lambda^{-n-1} T^n. \quad (8.2)$$

This is a support-certified Neumann inverse: the scalar λ^{-1} has positive valuation. It belongs to $\mathcal{A}_\Gamma(H)$, irrespective of the ordinary growth of $\|T^n\|$.

Finite scalar with nonspectral residue. Write $\lambda = a + \varepsilon$, where $a \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_K$. If $a \notin \sigma_{\mathbb{C}}(T)$, let $B = (T - aI)^{-1} \in \mathcal{B}(H)$. Then

$$(T - aI - \varepsilon I)^{-1} = \sum_{n \geq 0} \varepsilon^n B^{n+1}. \quad (8.3)$$

The formula includes $\varepsilon = 0$. For nonzero ε , use lemma 2.3 on εB ; coefficients need not satisfy any ordinary norm-convergence estimate.

Undisplaced ordinary spectral point. If $\varepsilon = 0$ and $a \in \sigma_{\mathbb{C}}(T)$, then $T - aI$ is not an ordinary bijection, by the bounded inverse theorem. A nonzero ordinary kernel vector remains a Hahn kernel vector. If there is no kernel but surjectivity fails, choose a constant f outside its ordinary range. The coefficient at zero in any putative solution would have to solve $(T - aI)y_0 = f$, which is impossible. Thus the Hahn operator is not bijective.

Displaced isolated point. Suppose a is isolated and $\varepsilon \neq 0$. Use lemma 8.1; let P, Q project onto N, M . The inverse is

$$(T - aI - \varepsilon I)^{-1} = -\varepsilon^{-1}P + \sum_{n \geq 0} \varepsilon^n S^{-n-1}Q. \quad (8.4)$$

The first term is a single scalar Hahn series times a bounded projection; the second is a positive-order Neumann series on M . Both have well-ordered operator support and lie in $\mathcal{A}_\Gamma(H)$. The finite orthogonal splitting verifies the inverse on both summands. If $M = 0$, the second summand is simply zero.

Displaced accumulation point. Suppose $a \in \sigma_{\mathbb{C}}(T)'$ and $\varepsilon \neq 0$. On M in lemma 8.1, the ordinary S is injective and not onto. The perturbation $-\varepsilon I$ has strictly positive order. By theorem 7.1, $S - \varepsilon I$ is not onto $M((t^\Gamma))$. The constant decomposition $H = N \oplus M$ extends to a decomposition of Hahn spaces, and the full operator is block diagonal. It is therefore not onto either. Explicitly, a constant $f \in M \setminus \text{Ran } S$ cannot be its M -component output.

These cases prove exactly the stated set formula. In every nonspectral case we constructed an inverse in $\mathcal{A}_\Gamma(H)$; in every spectral case we proved failure of algebraic bijectivity. This proves equality of the two spectral definitions as well. $\square$

8.3 Point spectrum is rigid, despite spectral thickening

Theorem 8.3 (No new normal eigenvalues). *For ordinary bounded normal T ,*

$$\sigma_{p,\Gamma}(T) = \sigma_{p,\mathbb{C}}(T), \quad \ker(T - aI) = \ker_H(T - aI)((t^\Gamma)) \quad (a \in \mathbb{C}).$$

In particular, the new nonreal spectral points of a self-adjoint operator are not eigenvalues.

Proof. Infinite scalars are in the resolvent by theorem 8.2. Suppose $(T - aI - \varepsilon I)x = 0$ with $x \neq 0$ and $\varepsilon \in \mathfrak{m}_K$. At its leading exponent δ , one obtains $(T - aI)x_\delta = 0$. Thus a is an ordinary eigenvalue and $x_\delta \in N = \ker_H(T - aI)$. If $\varepsilon \neq 0$, apply the ordinary orthogonal projection P onto N . Normality implies $P(T - aI) = 0$, so the full equation gives $-\varepsilon Px = 0$. Hence $Px = 0$, contradicting $Px_\delta = x_\delta \neq 0$. Therefore $\varepsilon = 0$. For a constant a , the kernel equation is coefficientwise, yielding the claimed eigenspace identity. $\square$

Example 8.4 (Normality is indispensable). Let B be the backward shift on $\ell^2(\mathbb{N}_0)$: $Be_0 = 0$, $Be_n = e_{n-1}$ for $n \geq 1$. For $s = t^\eta$,

$$x = \sum_{n \geq 0} s^n e_n \in \mathcal{H}_\Gamma, \quad Bx = sx.$$

The support $\{n\eta : n \geq 0\}$ is well ordered. This produces a new nonconstant eigenvalue for the nonnormal operator B , in direct contrast to theorem 8.3. No ordinary Hilbert convergence of the vector series is asserted or needed.

9 Consequences that separate finite and infinite spectral geometry

9.1 Self-adjoint spectra are real exactly in the finite-spectrum case

Corollary 9.1 (A sharp spectral-reality criterion). *Let T be ordinary bounded self-adjoint. The following are equivalent:*

- (i) $\sigma_{\Gamma}(T) \subseteq F$;
- (ii) $\sigma_{\mathbb{C}}(T)$ is finite;
- (iii) T has a finite ordinary orthogonal spectral resolution $T = \sum_{j=1}^r a_j P_j$ with $a_j \in \mathbb{R}$.

There is no finite-rank hypothesis on the projections P_j .

Proof. The usual self-adjoint spectrum is a nonempty compact subset of $\mathbb{R}$. A finite set has no accumulation points, so theorem 8.2 proves (ii) $\Rightarrow$ (i). An infinite compact subset has an accumulation point a . For any $s = t^\eta$, $\eta > 0$, the scalar $a + is$ belongs to $\sigma_{\Gamma}(T)$ and does not belong to F , proving the converse. The equivalence with (iii) is the ordinary spectral theorem applied to a finite spectral set. $\square$

The criterion is finite *spectrum*, not finite dimension. For example, an infinite-rank orthogonal projection with infinite-dimensional kernel has spectrum exactly $\{0, 1\}$ even after Hahn extension. Infinite multiplicity at an isolated point does not create a monad. The accumulation set in (8.1) therefore cannot be replaced by an essential spectrum that includes such points.

9.2 Ordinary compact operators and a continuous-spectrum example

Corollary 9.2 (Ordinary compact normal extension). *If T is an ordinary compact normal operator of infinite rank, then*

$$\sigma_{\Gamma}(T) = \mathfrak{m}_K \cup \{\text{nonzero ordinary eigenvalues of } T\}.$$

If its ordinary spectrum is finite, there is no thickening.

Proof. The ordinary compact normal spectral theorem says that nonzero spectral points are isolated eigenvalues of finite multiplicity and zero is the only possible accumulation point. Infinite rank forces zero to be an accumulation point. Apply theorem 8.2. $\square$

Here “compact” describes the original operator on the ordinary Hilbert space. It makes no claim of compactness or compactoidness in the Hahn valuation topology. The sequence (De_n) for $De_n = n^{-1}e_n$, for instance, has pairwise valuation distance one, despite $\|De_n\|_H \rightarrow 0$ in the ordinary norm.

For another concrete example, let T be multiplication by the coordinate on $L^2([0, 1], \mathbb{C})$. Its ordinary spectrum is $[0, 1]$, it has no eigenvalues, and every point of its spectrum is an accumulation point. Thus

$$\sigma_{\Gamma}(T) = [0, 1] + \mathfrak{m}_K, \quad \sigma_{p,\Gamma}(T) = \emptyset.$$

The usual facts here can be seen directly: away from $[0, 1]$ the reciprocal multiplier is bounded; near any point of $[0, 1]$, vectors supported in arbitrarily small intervals prevent a bounded

inverse; a singleton has measure zero and cannot support a nonzero eigenvector. This gives full infinitesimal fibers above a continuous real interval, without adding a single eigenvector.

9.3 Enlarging the scalar workspace does not repair the defect

Corollary 9.3 (Persistence under ordered-group extension). *Let $\Gamma \subseteq \Delta$ be ordered divisible groups and use the natural coefficient-preserving embedding $K_\Gamma \subseteq K_\Delta$. For constant normal T ,*

$$\sigma_\Delta(T) \cap K_\Gamma = \sigma_\Gamma(T).$$

Moreover, the nonsurjectivity in corollary 7.2 persists after such an extension for the same S and E .

Proof. An element of K_Γ has the same leading valuation, residue and infinitesimal status in the larger group. Apply the exact spectral formula. For the second assertion use the same complement W in theorem 7.1, now forming $W((t^\Delta))$. $\square$

This is a genuine surcomplex-workspace statement. Adding further surreal scales does not turn a prohibited leading ordinary Hilbert coefficient into an allowed one. It does not assert that an unspecified larger class-sized notion of vector could never behave differently. Proposition 25.2 is the corresponding naturality statement for Part II; it is about a different construction and is proved separately.

10 Which constant data have an infinitesimal resolvent?

10.1 The inverse-smooth domain

Even when $S - sI$ is not onto, some constant right-hand sides have a unique solution. The exact criterion is stronger than belonging merely to $\text{Ran } S$.

Theorem 10.1 (All inverse moments are necessary and sufficient). *Let $S : V \rightarrow V$ be injective and $s = t^\eta$ with $\eta > 0$. For constant $f \in V$, the equation*

$$(S - sI)y = f, \quad y \in V((t^\Gamma)), \tag{10.1}$$

has a solution if and only if

$$f \in \bigcap_{k \geq 1} \text{Ran } S^k. \tag{10.2}$$

In that case the solution is unique and is

$$y = \sum_{n \geq 0} s^n S^{-n-1} f. \tag{10.3}$$

The inverse powers denote the unique preimages, not an everywhere-defined inverse operator on V .

Proof. Sufficiency follows by coefficientwise telescoping of (10.3); its support is contained in the well-ordered set $\{n\eta : n \geq 0\}$. Injectivity follows from theorem 7.1.

For necessity, partition the support of a proposed solution according to cosets of $\mathbb{Z}\eta$ in Γ . Restricting a Hahn series to a coset preserves well-ordering, and both S and multiplication by s preserve that coset. On a coset other than $\mathbb{Z}\eta$, the restricted equation is homogeneous. Its least nonzero coefficient would be killed by S , which is impossible. Thus the support lies in $\mathbb{Z}\eta$.

A well-ordered subset of $\mathbb{Z}\eta$ is bounded below in its integer index. For $f \neq 0$, the valuation-preservation assertion of theorem 7.1 forces the least index to be zero. Write $y = \sum_{n \geq 0} s^n y_n$, allowing zero coefficients. The equation gives

$$Sy_0 = f, \quad Sy_n = y_{n-1} \quad (n \geq 1).$$

Consequently $S^{n+1}y_n = f$ for every n , proving (10.2) and the formula. The zero-data case is immediate. $\square$

Example 10.2 (The diagonal inverse-smooth space). For $De_n = n^{-1}e_n$ on $H = \ell^2(\mathbb{N}_{\geq 1})$, the admissible constant data are exactly

$$H_\infty = \left\{ f = (f_n) : \sum_{n \geq 1} n^{2k} |f_n|^2 < \infty \text{ for every integer } k \geq 1 \right\}. \quad (10.4)$$

The formula follows because $D^{-k}f = (n^k f_n)$. Finitely supported vectors belong to H_∞ , so this subspace is dense in the *ordinary* Hilbert topology. Yet $f_n = n^{-2}$ does not belong to it, although $f \in \text{Ran } D$. Thus even some data with an ordinary first preimage lack an infinitesimal resolvent.

The same statement for $D - isI$ follows by replacing the recurrence with $Dy_n = iy_{n-1}$. Its solution, when it exists, is $\sum_{n \geq 0} i^n s^n D^{-n-1}f$. No uniform bound on the ordinary norms of these coefficients is required, only that each one actually belongs to H .

10.2 Why individual scalar inverses are insufficient

Every diagonal scalar $n^{-1} - s$ is invertible in K and has expansion

$$(n^{-1} - s)^{-1} = \sum_{k \geq 0} n^{k+1} s^k.$$

But applying these inverses coordinatewise to f requires the entire coefficient vector $(n^{k+1}f_n)_n$ to lie in ℓ^2 at each k . The Hahn vector space imposes this condition before any comparison of higher scales. There is therefore no contradiction between invertibility of every diagonal entry and failure of surjectivity of the infinite operator. A putative coordinatewise inverse has left the chosen vector space, rather than found a missing scalar inverse. In Part II, where the vector space and the operator algebra are different, the corresponding coordinatewise inverse is admissible; section 3.2 compares the two situations directly.

11 Coercive operators with continuum-dimensional defects

11.1 An explicit large ordinary cokernel

Continue with $H = \ell^2(\mathbb{N}_{\geq 1})$ and $De_n = n^{-1}e_n$. The operator D is bounded, positive, self-adjoint and injective. It is not onto because $(n^{-1})_n \in \ell^2$ has no preimage in ℓ^2 .

Lemma 11.1 (Continuum many independent defect vectors). *For $r \in (1/2, 3/2]$, let $f^{(r)} = (n^{-r})_{n \geq 1} \in H$. Their classes in $H/\text{Ran } D$ are linearly independent over $\mathbb{C}$.*

Proof. Take distinct $r_1 < \dots < r_k$ and complex coefficients with $c_1 \neq 0$. For $g = \sum_{j=1}^k c_j f^{(r_j)}$, one has

$$ng_n = c_1 n^{1-r_1} (1 + o(1)).$$

Its squared magnitude is bounded below by a positive real multiple of n^{2-2r_1} for sufficiently large n . Since $r_1 \leq 3/2$, the exponent is at least -1 , and the series of those squared magnitudes diverges. Thus $(ng_n) \notin \ell^2$, which is exactly $g \notin \text{Ran } D$. There are continuum many choices of r . $\square$

Theorem 11.2 (Every positive-order perturbation retains a large defect). *For every $E \in \mathcal{B}(H)((t^\Gamma))$ of positive order, including zero,*

$$\dim_K \operatorname{coker}(D + E) \geq 2^{\aleph_0}.$$

The same lower bound holds with leading operator D^2 .

Proof. Apply corollary 7.3 and lemma 11.1. Since $\operatorname{Ran} D^2 \subseteq \operatorname{Ran} D$, the same vectors remain independent modulo $\operatorname{Ran} D^2$. $\square$

11.2 A nonreal spectral point with a strong lower bound

For $s = t^\eta$, $\eta > 0$, set $A = D - isI$. Since $D = D^*$,

$$A^*A = D^2 + s^2I, \quad \|Ax\|^2 = \|Dx\|^2 + s^2\|x\|^2 \geq s^2\|x\|^2. \quad (11.1)$$

Nevertheless, A is not onto by theorem 7.1. Thus the usual lower-bound argument for excluding nonreal self-adjoint spectrum stops precisely at surjectivity, not at injectivity.

The operator even has an attained upper norm bound:

$$\|Ax\| \leq \sqrt{1 + s^2} \|x\|, \quad \|Ae_1\| = \sqrt{1 + s^2}. \quad (11.2)$$

Indeed D is an ordinary contraction and its extension remains a contraction; for this diagonal operator the inequality also follows by writing $I - D^2 = R^*R$ with ordinary bounded diagonal $Re_n = \sqrt{1 - n^{-2}}e_n$. Equations (11.1)–(11.2) show that neither lack of a bound nor lack of an attained norm explains the failure of inversion.

11.3 Positive coercivity does not imply surjectivity

Theorem 11.3 (A coercive noninvertible self-adjoint operator). *Let*

$$C = D^2 + s^2I. \quad (11.3)$$

Then C is everywhere-defined, adjointable and self-adjoint, with

$$\langle Cx, x \rangle \geq s^2 \langle x, x \rangle, \quad (11.4)$$

$$\|Cx\| \leq (1 + s^2)\|x\|. \quad (11.5)$$

The upper bound is attained at e_1 . The operator is injective but not surjective, and its cokernel has dimension at least $2^{\aleph_0}$ over K . Specifically, the constant vector $f = (n^{-2})_{n \geq 1}$ is not in its range.

Proof. Self-adjointness and (11.4) follow from $\langle D^2x, x \rangle = \langle Dx, Dx \rangle \geq 0$. The contraction property of D^2 , the triangle inequality and $Ce_1 = (1 + s^2)e_1$ give (11.5) and its attainment. The leading coefficient operator D^2 is injective, so theorem 7.1 gives injectivity, and theorem 11.2 gives the cokernel bound.

For a direct witness, if $Cy = f$, valuation preservation forces $v(y) = 0$. The coefficient at zero must solve $D^2y_0 = f$, that is, $(y_0)_n = 1$ for every n . This is not an ℓ^2 vector. Thus no such y exists. $\square$

For clarity, coercivity is a bound by a strictly positive element s^2 of the ordered scalar field. It is not a bound by a positive *ordinary real* independent of the infinitesimal scale. If instead $C \geq cI$ with ordinary $c > 0$ and an ordinary bounded positive leading operator with this bound, the classical inverse and the positive-order Neumann construction would apply. The infinitesimal scale is exactly where the distinction becomes mathematically significant.

Corollary 11.4 (Failure of a specified Lax–Milgram analogue). *On this Hahn inner-product space, a bounded coercive sesquilinear form need not solve all variational problems, even when the right-hand side is represented by an actual vector. In particular,*

$$b(u, v) = \langle Cu, v \rangle, \quad \ell(v) = \langle f, v \rangle, \quad f = (n^{-2})_n,$$

defines a bounded coercive form and a bounded conjugate-linear functional, but there is no u satisfying $b(u, v) = \ell(v)$ for all $v \in \mathcal{H}_\Gamma$.

Proof. Cauchy–Schwarz and (11.5) bound the form; (11.4) is coercivity, and $|\ell(v)| \leq \|f\| \|v\|$. If the identity held for all v , nondegeneracy of the inner product would give $Cu = f$, contradicting theorem 11.3. $\square$

This does not contradict the ordinary complex Lax–Milgram theorem: the scalar field, the topology and the duality structure have changed. It is an explicit obstruction to transferring that theorem using only its familiar-looking algebraic inequalities.

12 Completeness does not restore orthogonal projection

12.1 The rank-one topology and its completeness

In this section only, assume Γ is a nonzero divisible ordered subgroup of $\mathbb{R}$. For $x \neq 0$ define

$$\rho(x) = e^{-v(x)}, \quad \rho(0) = 0, \quad d(x, y) = \rho(x - y).$$

The scalar absolute value here is $|a|_v = e^{-v(a)}$, not the F -valued modulus. The metric is ultrametric. Its topology agrees with the topology of the field-valued norm balls on $\mathcal{H}_\Gamma$: a sufficiently higher valuation makes a vector smaller than any prescribed positive radius, and choosing a still higher monomial radius gives the converse inclusion of neighborhood bases.

It is not the ordinary Hilbert topology on the constant copy of H . Every nonzero constant vector has ρ -size one. Nor is it the fine subspace topology from the full surcomplex class, where the set-sized scalar workspace is discrete [1].

Proposition 12.1 (Rank-one valuation completeness). *For any complex vector space V , the metric space $V((t^\Gamma))$ with metric d is complete. In particular, $\mathcal{H}_\Gamma$ is complete in this topology.*

Proof. Let (x_n) be Cauchy. For each real cutoff R , all sufficiently late x_n have exactly the same coefficients at exponents $\gamma \leq R$, since eventually $v(x_n - x_m) > R$. Define x_γ to be the eventually stable coefficient.

The support of this coefficient family is well ordered. Otherwise it would contain a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$. Choose a cutoff at least γ_1 . Below that cutoff the entire stable family agrees with one sufficiently late Hahn vector x_N , whose well-ordered support cannot contain that sequence. This is a contradiction. Thus $x = \sum x_\gamma t^\gamma$ belongs to $V((t^\Gamma))$. By the same stabilization statement, $v(x_n - x) > R$ eventually for every real R , proving convergence. Notice that no coefficient-space topology was used: stabilization is exact, not approximate. $\square$

12.2 A proper closed subspace with zero orthogonal complement

Theorem 12.2 (Closed-range orthogonality failure). *For the coercive $C = D^2 + s^2 I$, its range $M = C\mathcal{H}_\Gamma$ is closed and proper in the complete rank-one space $\mathcal{H}_\Gamma$, but*

$$M^\perp = \{0\}, \quad M^{\perp\perp} = \mathcal{H}_\Gamma \neq M. \tag{12.1}$$

In particular, M is not the range of a self-adjoint idempotent.

Proof. By theorem 7.1, $v(Cx) = v(x)$; hence C is an isometry for ρ . The image of a complete metric space under an isometry is complete and therefore closed in any metric ambient space. It is proper by theorem 11.3.

If $z \in M^\perp$, then $\langle Cx, z \rangle = 0$ for every x . Since $C = C^*$, $\langle x, Cz \rangle = 0$ for every x , so $Cz = 0$. Injectivity gives $z = 0$. This proves (12.1). A self-adjoint idempotent P has $\mathcal{H}_\Gamma = \text{Ran } P \oplus \ker P$ and $\ker P = (\text{Ran } P)^\perp$, by the finite algebraic projection identities. If its range were M , the complement would be zero and the range all of $\mathcal{H}_\Gamma$, a contradiction. $\square$

The displayed range is not orthogonally closed, although it is metrically closed. Thus the usual Hilbert identification of these two closure notions fails. Known non-Archimedean failures of orthomodularity also warn against unconditional projection theorems [8, 9]. The theorem above gives a particularly rigid witness: it is the closed range of a bounded, coercive, self-adjoint injection and retains a continuum-sized algebraic deficit.

12.3 A continuous functional outside inner-product duality

Proposition 12.3 (Explicit failure of valuation Riesz representation). *There is a nonzero K -linear, valuation-continuous functional $F : \mathcal{H}_\Gamma \rightarrow K$ that vanishes on $M = C\mathcal{H}_\Gamma$ but is not of the form $F(x) = \langle x, z \rangle$ for any $z \in \mathcal{H}_\Gamma$.*

Proof. Choose an algebraic complement W of $\text{Ran } D^2$ in H , and take the projection Π in theorem 7.1 for $S = D^2$, $E = s^2I$. Its coefficients have nonnegative exponents: U and S are constant, E has positive order, and $(I + UE)^{-1}$ has nonnegative order. Consequently $v(\Pi x) \geq v(x)$. Choose a nonzero complex linear functional $\psi : W \rightarrow \mathbb{C}$, and extend it coefficientwise to $\psi_\Gamma : W((t^\Gamma)) \rightarrow K$. Set $F = \psi_\Gamma \Pi$. Coefficient extraction gives $v(Fx) \geq v(x)$, so F is valuation-continuous. It is nonzero because Π is the identity on $W((t^\Gamma))$, and it kills M . If $F(x) = \langle x, z \rangle$, then vanishing on M implies $z \in M^\perp = 0$, which would force $F = 0$. $\square$

The algebraic maps U and ψ need not be continuous for the ordinary Hilbert norm. That does not prevent the displayed Hahn maps from being valuation-continuous. Confusing those topologies would incorrectly apply an ordinary closed graph or Riesz argument. Conversely, this proposition does not contradict theorem 6.1: the functional just constructed has no adjoint represented by a Hahn vector.

13 When does an operator have a least field-valued norm?

13.1 Bounds versus the existence of their least element

For $A \in \mathcal{A}_\Gamma(H)$, let

$$\mathcal{U}(A) = \{M \in F_{\geq 0} : \|Ax\| \leq M\|x\| \text{ for all } x \in \mathcal{H}_\Gamma\}.$$

Corollary 6.2 makes this set nonempty. By a field-valued *operator norm* we mean a least element of this set, not an arbitrary chosen bound and not the valuation norm of the operator.

Lemma 13.1 (Ordinary contraction bounds survive). *If $T \in \mathcal{B}(H)$ and $n = \|T\|_H$, then $\|Tx\| \leq n\|x\|$ for every $x \in \mathcal{H}_\Gamma$. For nonzero x with leading coefficient u , the finite scalar $\|Tx\|/\|x\|$ has standard part*

$$\text{st}\left(\frac{\|Tx\|}{\|x\|}\right) = \frac{\|Tu\|_H}{\|u\|_H}.$$

The right side is zero if $Tu = 0$.

Proof. The ordinary positive operator $n^2I - T^*T$ has a bounded positive square root R by ordinary Hilbert spectral theory. The coefficientwise identity $R^*R = n^2I - T^*T$ implies $n^2\|x\|^2 - \|Tx\|^2 = \|Rx\|^2 \geq 0$. For the standard part, use the leading norm formula in proposition 5.1. If $Tu = 0$, the output has strictly larger valuation or is zero, so the quotient has residue zero. $\square$

Theorem 13.2 (Norm-attainment criterion). *For an ordinary bounded operator T on nonzero H , the set $\mathcal{U}(T)$ has a least element if and only if T attains its ordinary operator norm on a unit vector. When it does, the least element is $\|T\|_H$ and is attained by a constant unit vector in $\mathcal{H}_\Gamma$.*

Proof. Put $n = \|T\|_H$. If T attains n , the constant maximizing vector forces every upper bound to be at least n , while lemma 13.1 proves that n is a bound. This includes $T = 0$, which attains its norm on every unit vector.

Now suppose T does not attain its norm; then $n > 0$. For each nonzero $x \in \mathcal{H}_\Gamma$, lemma 13.1 gives

$$\text{st}\left(\frac{\|Tx\|}{\|x\|}\right) < n.$$

Therefore every positive finite M with $\text{st}(M) = n$ is an upper bound, including $n - \varepsilon$ for every positive infinitesimal ε . Every finite M with residue greater than n and every positive infinite M is a bound as well.

No finite nonnegative M with residue less than n can be a bound: choose an ordinary unit vector u with $\|Tu\|_H > \text{st}(M)$ using the definition of the ordinary supremum. Its constant extension violates that bound. We have thus described all upper bounds. A bound with residue greater than n , or an infinite bound, can be decreased to n . A bound with residue n can be decreased by any fixed positive infinitesimal and remains positive with the same residue. There is no least upper bound. $\square$

Example 13.3 (A bounded self-adjoint operator without a field-valued norm). Let $Te_n = (1 - 1/(n+1))e_n$ on $\ell^2(\mathbb{N}_{\geq 1})$. Its ordinary norm is one but is not attained. Indeed, for every nonzero u ,

$$\|u\|_H^2 - \|Tu\|_H^2 = \sum_{n \geq 1} \left(1 - \left(1 - \frac{1}{n+1}\right)^2\right) |u_n|^2 > 0,$$

where this is an ordinary convergent real sum. The diagonal values approach one, proving the supremum is one. Hence its Hahn extension has many field-valued bounds, including $1 - s$ and $1 - 2s$, but no least one.

13.2 The two mechanisms are independent

The operator $D^2 + s^2I$ in theorem 11.3 has an attained least upper norm bound $1 + s^2$ but is not invertible. The operator in example 13.3 fails norm attainment for a different reason: every individual Hahn vector has a strictly submaximal *ordinary leading ratio*, so infinitesimally smaller bounds cannot be distinguished by any one vector. Neither phenomenon should be used as an explanation for the other.

14 Why finite invertibility is misleading

Let $D_N = \text{diag}(1, 1/2, \dots, 1/N)$ and $C_N = D_N^2 + s^2I$. Every C_N is invertible over K . For the truncated right-hand side $f_N = (n^{-2})_{1 \leq n \leq N}$,

$$(C_N^{-1}f_N)_n = \frac{1}{1 + n^2s^2} = \sum_{k \geq 0} (-1)^k n^{2k} s^{2k}. \quad (14.1)$$

The constant coefficient of this finite solution is the N -vector of ones, whose ordinary Hilbert norm is $\sqrt{N}$. No single ordinary ℓ^2 vector can serve as the stable constant coefficient as $N \rightarrow \infty$. This directly identifies the obstruction missed by every finite matrix determinant.

For $A_N = D_N - isI$, all diagonal resolvents likewise exist, with

$$(n^{-1} - is)^{-1} = \sum_{k \geq 0} i^k n^{k+1} s^k.$$

The leading inverse coefficient has ordinary norm N . Infinite extension would require the unbounded ordinary operator $\text{diag}(n)$ as a bounded coefficient, contradicting theorem 6.1. More strongly, particular constant data fail the vector-space condition of theorem 10.1; thus the problem is actual nonsurjectivity, not just exclusion from a preferred inverse algebra.

Remark 14.1 (The same matrix in the other part). The operator $\text{diag}(n)$ excluded here is admitted in Part II, where the coefficient algebra is $\text{RCF}_I(\mathbb{C})$ rather than $\mathcal{B}(H)$, and where the finite-section question has a positive answer of a different kind: theorem 21.4 gives an exact finite spatial certificate for each individual coefficient. The two statements are about different algebras and do not conflict; see section 3.2.

15 Part I: interpretation and remaining boundaries

15.1 What has been settled in this construction

The questions raised by this particular scalar-extension model now have sharp answers. Adjointability is not an opaque class condition: it is exactly global Hahn support with bounded ordinary coefficients. For constant normal operators, infinitesimal resolvent behavior is completely classified by the ordinary spectral accumulation set. Infinitesimal perturbations of injective leading operators cannot remove any ordinary range defect. The leading ordinary coefficient gives a direct and robust witness, even where every finite section is invertible.

These statements distinguish spectral membership from eigenvalue existence, compactness in the original Hilbert topology from compactness in the valuation topology, upper norm bounds from a least such bound, and metric completeness from orthogonal completeness. None of those pairs can be silently identified after extending the scalars.

15.2 What would require a different theorem

The formula (8.1) concerns *constant normal* operators. General Hahn operators with multiple coefficient scales can have more complicated spectra; example 8.4 already shows why the point-spectrum rigidity fails without normality. The perturbation theorem handles injective leading coefficients and proves a large class of obstructions, but is not a classification of all spectra of all Hahn operator series.

One could change the vector construction to allow coefficientwise objects outside H , choose a different sequence-space completion, or impose a stronger orthogonal-completeness axiom. Such changes must be specified and analyzed rather than treated as harmless notational variations. Enlarging only the exponent group is not enough, by corollary 9.3.

15.3 Why this matters beyond a counterexample

In finite dimension, replacing $\mathbb{R}/\mathbb{C}$ by a real-closed field and its complexification preserves much of spectral algebra. In infinite dimension, that replacement cannot by itself preserve the analytic role of Hilbert space. An infinitesimal imaginary displacement may fail to produce a resolvent, even when it gives a strict lower norm bound. An infinitesimal positive regularizer may leave a

continuum of unsolved linear equations. These are precise obstructions to using the model as an automatic replacement for complex Hilbert space, not a rejection of surcomplex analysis in every possible formulation.

The constructive side is equally useful. The automatic adjoint theorem gives a well-defined algebra for exact symbolic manipulation. The inverse formulas identify every ordinary normal resolvent that actually exists. The inverse-smooth criterion identifies exactly which constant data remain solvable at a monomial infinitesimal shift. The defect splitting gives explicit noncanonical projections and an exact algebraic quotient. Together these form a workable theory with both positive structure and sharp limits.

Part II. Row- and column-finite Hahn matrices: exact synthesis and algebra-relative spectra

Standing hypotheses for Part II. Γ is a set-sized ordered abelian group, *not assumed divisible*; I is a nonempty index set. “Spectrum” in this part means $\sigma_{\mathcal{A}_I}$, failure of two-sided invertibility inside the named unital K -algebra $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$, acting on $\mathcal{H}_I = \mathbb{C}^{(I)}((t^\Gamma))$. It is not a Banach-algebra spectrum, and it is not the failure-of-bijectivity spectrum of Part I, which lives on a different vector space.

16 The question, the answer, and its scope

Passing from finitely many to infinitely many coordinates changes the problem: products can cease to be defined, formal eigenvectors can escape the chosen vector space, and infinitely many small denominators need not define a bounded operator.

We address the following precise question. Suppose an infinite diagonal matrix has distinct ordinary real entries and is perturbed by genuinely infinitesimal surcomplex coefficients. Is there a natural associative operator algebra, together with a vector space carrying a positive surreal-valued inner product, in which *all* eigenvectors can be constructed coherently, summed against arbitrary vectors, and used to recover exact spectral projections and resolvents?

The answer is affirmative in a coefficientwise locally finite Hahn category. The category is part of the theorem, not an incidental convention. The operators need not be bounded on an ordinary Hilbert space; the vector space is not ordinary ℓ^2 ; and Hahn summation is not shorthand for a sequence of topological limits. Those distinctions allow accumulation of real energy levels while retaining coefficientwise exactness.

The repository was checked for this part at commit `a3124af`; the complete commit identifier appears in the bibliography [3] and in appendix B. In particular, a still-unfinished Lean implementation of a normal-form bridge is not being treated as an open mathematical conjecture. The surcomplex interpretation in section 25 uses the established normal-form theorem [4], not an unproved declaration in a software project.

17 The coefficient algebra and its Hahn vector space

17.1 Row- and column-finite coefficients

Let I be a nonempty set. A matrix $M = (M_{ij})_{i,j \in I}$ is *row- and column-finite* if each of its rows and each of its columns has finite support. The number of nonzero entries need not be uniformly bounded as the row or column varies. Set

$$R_I = \text{RCF}_I(\mathbb{C}), \quad V_I = \mathbb{C}^{(I)}.$$

Every diagonal matrix with entries in $\mathbb{C}$ belongs to R_I , even when its entries are unbounded in the ordinary absolute value.

Lemma 17.1. *The usual finite row-by-column multiplication makes R_I an associative unital $\mathbb{C}$ -algebra. Conjugate transpose is an involution, and R_I acts faithfully on V_I . With the ordinary finite-support inner product, $\langle Mx, y \rangle = \langle x, M^*y \rangle$.*

Proof. For a fixed row of MN , only the finitely many rows of N selected by that row of M can contribute. Their supports have finite union. The analogous column argument proves column-finiteness. Every entry in an iterated product involves only finitely many intermediate indices, so associativity follows from distributivity of finite sums. The identity matrix belongs to R_I , and conjugate transpose interchanges the two finiteness conditions. A finite-support vector is taken to a finite-support vector because the relevant columns have finite union of supports. The adjoint identity is a finite computation. Finally, an operator acting as zero on every standard vector e_i has every column zero. $\square$

The requirement is deliberately imposed on *coefficients*, rather than on the complete matrix of Hahn entries. The eventual diagonalizer will generally have infinitely many nonzero entries in a row or column, spread over increasingly complicated exponents.

17.2 The Hahn algebra and module

With the conventions of section 2.1, define

$$K = \mathbb{C}((t^\Gamma)), \quad F = \mathbb{R}((t^\Gamma)), \quad \mathcal{A}_I = R_I((t^\Gamma)), \quad \mathcal{H}_I = V_I((t^\Gamma)). \quad (17.1)$$

The usual Hahn convolution makes K a field and F an ordered subfield; a nonzero real series is positive exactly when its leading coefficient is positive. The field property does not require Γ to be divisible: a nonzero series factors as a nonzero constant times a monomial times a unit of the form $1 + \text{positive support}$, whose inverse is given by lemma 2.3. Write $\mathfrak{m}_F = \{f \in F : \text{supp } f \subset \Gamma_{>0}\}$; it includes zero and is not the cone of positive-valued elements.

Proposition 17.2. *The space $\mathcal{A}_I$ is an associative unital K -algebra with involution*

$$\left(\sum M_\gamma t^\gamma \right)^* = \sum M_\gamma^* t^\gamma.$$

It acts faithfully on the K -vector space $\mathcal{H}_I$ by convolution. For products in this algebra and its module,

$$v(XY) \geq v(X) + v(Y), \quad v(Xx) \geq v(X) + v(x).$$

The order function on $\mathcal{A}_I$ is not asserted to be a field valuation: $\mathcal{A}_I$ has zero divisors.

Proof. At a prescribed exponent, lemma 2.1(i) leaves finitely many coefficient products; lemma 17.1 puts each in R_I or V_I . The same finite-factor support argument permits reassociation of every triple product. Scalar Hahn series act through scalar matrices. The support of a product is contained in the sum of the supports, proving the displayed inequalities. Faithfulness follows by evaluating on each e_i and then comparing coefficients. The adjoint identity follows coefficientwise. $\square$

17.3 The inner product and its positivity

Proposition 17.3. *The formula*

$$\langle x, y \rangle = \sum_{\alpha, \beta} \langle x_\alpha, y_\beta \rangle_{\mathbb{C}(t)} t^{\alpha+\beta} = \sum_{i \in I}^H \overline{x_i} y_i \quad (17.2)$$

defines a conjugate-linear/linear K -valued inner product on $\mathcal{H}_I$. For $x \neq 0$,

$$\langle x, x \rangle \in F_{>0}, \quad v(\langle x, x \rangle) = 2v(x).$$

*The adjoint relation $\langle Xx, y \rangle = \langle x, X^*y \rangle$ holds for $X \in \mathcal{A}_I$.*

Proof. For each total exponent there are finitely many pairs (α, β) , and each pair involves finite-support vectors. Thus the family over i in (17.2) satisfies both conditions for strong Hahn summability. Sesquilinearity and the adjoint relation follow from finite coefficient calculations. If $\gamma = v(x)$, the coefficient at 2γ in $\langle x, x \rangle$ is

$$\sum_i |(x_\gamma)_i|^2 > 0.$$

There can be no smaller exponent, and no other pair of support exponents sums to 2γ . This proves positivity and the valuation formula. $\square$

Remark 17.4 (No divisibility, and no norm). Proposition 17.3 is positive without assuming that every positive element of F has a square root. Indeed, for arbitrary Γ the field F need not be real closed. The only roots used in the main construction are roots of units with residue 1, obtained by binomial Hahn series through corollary 2.6. In particular no analogue of $\|x\| = \sqrt{\langle x, x \rangle}$ is introduced here; the norm of proposition 5.1 is a Part I object, but its construction does not require divisibility. The even-leading-exponent argument also supplies a square root of the squared length in Part II; we simply do not use a norm in this part's algebraic construction.

Lemma 17.5 (Strong sums are respected). *If (x_a) is strongly summable in $\mathcal{H}_I$ and $X \in \mathcal{A}_I$, then (Xx_a) is strongly summable and $X(\sum_a^H x_a) = \sum_a^H Xx_a$. The analogous statement holds for multiplication by a fixed element in $\mathcal{A}_I$ and for taking the inner product with a fixed vector.*

Proof. The union of resulting supports lies in the sum of two well-ordered sets. At any exponent there are finitely many pairs of coefficient exponents, and for each such pair only finitely many family members can contribute. The claimed identities are therefore identities between finite sums at each coefficient. $\square$

18 Exact infinite diagonalization

18.1 The divided commutator

Let $(d_i)_{i \in I}$ be pairwise distinct elements of $\mathbb{C}$, and put $D = \text{diag}(d_i)$. Define two coefficientwise linear maps on R_I and hence on $\mathcal{A}_I$:

$$\pi(M) = \text{diag}(M_{ii}), \quad \mathcal{R}_D(M)_{ij} = \begin{cases} M_{ij}/(d_i - d_j), & i \neq j, \\ 0, & i = j. \end{cases} \quad (18.1)$$

They preserve support, and they preserve the row- and column-finiteness of every coefficient. On off-diagonal matrices, $\mathcal{R}_D$ is the inverse of $X \mapsto [D, X]$.

No uniform bound for $\mathcal{R}_D$ in a real operator norm is claimed. Every individual denominator is a nonzero ordinary complex number and therefore has Hahn valuation zero. Arbitrarily small ordinary gaps do not change supports. This is exactly the point at which the pairwise-distinct hypothesis does its work, and remark 18.4 records what happens when it is absent.

18.2 A similarity theorem without self-adjointness

Theorem 18.1 (Coherent simple-residue diagonalization). *Let $D = \text{diag}(d_i) \in R_I$ have pairwise distinct complex entries, and let $B \in \mathcal{A}_I$ have positive support. There exist unique*

$$V = \mathbf{1} + W \in \mathbf{1} + (\mathcal{A}_I)_{>0}, \quad \pi(W) = 0, \quad \Lambda = D + \Delta,$$

with Δ diagonal of positive support, such that

$$(D + B)V = V\Lambda. \quad (18.2)$$

The matrix V is invertible in $\mathcal{A}_I$. If $S = \text{supp } B$, then

$$\text{supp } W, \text{ sup } \Delta \subseteq S^* \setminus \{0\}.$$

The construction uses only finite coefficient operations, the nonzero numbers $d_i - d_j$, and strong Hahn summation.

Proof. Introduce a formal bookkeeping variable h , distinct from the Hahn monomials, and seek

$$V(h) = \sum_{n \geq 0} h^n V_n, \quad \Lambda(h) = D + \sum_{n \geq 1} h^n \Delta_n, \quad V_0 = \mathbf{1}, \quad \pi(V_n) = 0 \ (n \geq 1).$$

At degree $n \geq 1$, the equation $(D + hB)V(h) = V(h)\Lambda(h)$ is

$$[D, V_n] + BV_{n-1} = \Delta_n + \sum_{k=1}^{n-1} V_{n-k} \Delta_k.$$

Thus the recursion is forced:

$$T_n = BV_{n-1} - \sum_{k=1}^{n-1} V_{n-k} \Delta_k, \tag{18.3}$$

$$\Delta_n = \pi(T_n) = \pi(BV_{n-1}), \tag{18.4}$$

$$V_n = -\mathcal{R}_D(T_n). \tag{18.5}$$

The second equality in (18.4) follows because multiplying an off-diagonal V_{n-k} by a diagonal matrix does not create diagonal entries. These formulas define every coefficient and prove the formal eigenidentity.

Inductively, V_n and Δ_n are homogeneous of degree n in B , with support contained in nS . Each coefficient is in R_I , by lemma 17.1 and the support preservation of (18.1). Corollary 2.5 therefore licenses the substitutions

$$V = \mathbf{1} + \sum_{n \geq 1}^H V_n, \quad \Delta = \sum_{n \geq 1}^H \Delta_n.$$

It also licenses substitution in the products in the formal equation, proving (18.2). The support claim follows at once. The inverse of V exists by corollary 2.6.

For uniqueness, suppose (V', Λ') is another pair with the stated normalizations. Then $T = V^{-1}V'$ belongs to $\mathbf{1} + (\mathcal{A}_I)_{>0}$ and satisfies $\Lambda T = T \Lambda'$. The diagonal equation is $(\lambda_i - \lambda'_i)T_{ii} = 0$. Since T_{ii} has residue 1, it is nonzero, so $\lambda_i = \lambda'_i$. For $i \neq j$, the element $\lambda_i - \lambda_j$ has nonzero residue $d_i - d_j$, and hence $T_{ij} = 0$. Thus T is diagonal. Finally $V_{ii} = V'_{ii} = 1$ gives $T_{ii} = 1$, proving $T = \mathbf{1}$ and the asserted uniqueness. $\square$

The bookkeeping construction does not assert that $h = 1$ is a permissible substitution in arbitrary formal power series. It is permissible here because the degree- n coefficient is supported in nS , and lemma 2.1(ii) supplies the missing summability proof.

18.3 Positivity turns the similarity into a unitary

Theorem 18.2 (Normalized infinite Hahn spectral theorem). *Assume now that every d_i is real, that the d_i are pairwise distinct, and that $B = B^* \in \mathcal{A}_I$ has positive support. With $A = D + B$, there is a unique pair (U, Λ) satisfying*

$$AU = U\Lambda, \quad U^*U = UU^* = \mathbf{1}, \quad U \in \mathbf{1} + (\mathcal{A}_I)_{>0}, \tag{18.6}$$

where $\Lambda = \text{diag}(\lambda_i)$, $\lambda_i \in F$, $\lambda_i - d_i \in \mathfrak{m}_F$, and every diagonal entry U_{ii} is real and strictly positive. Both $U - \mathbf{1}$ and $\Lambda - D$ are supported in $S^* \setminus \{0\}$, where $S = \text{supp } B$.

Here $\mathfrak{m}_F$ includes zero and is not the cone of positive-valued elements. In particular, the correction to an eigenvalue may have either sign.

Proof. Use theorem 18.1 and write $v_i = Ve_i$. Since $Av_i = \lambda_i v_i$ and $A = A^*$,

$$\lambda_i \langle v_i, v_i \rangle = \langle v_i, Av_i \rangle = \overline{\langle v_i, Av_i \rangle}.$$

The norm square is positive and nonzero by proposition 17.3; therefore $\lambda_i = \overline{\lambda_i}$. For distinct i, j , self-adjointness similarly gives

$$(\lambda_j - \lambda_i) \langle v_i, v_j \rangle = 0.$$

The eigenvalue difference has nonzero real residue, so the columns are orthogonal.

Consequently $N = V^*V$ is diagonal. Because $V = \mathbf{1} + W$ and $W_{ii} = 0$,

$$N_{ii} = 1 + \langle We_i, We_i \rangle, \quad N = \mathbf{1} + \pi(W^*W). \quad (18.7)$$

The diagonal entries are positive with residue 1. The binomial series

$$H = N^{-1/2} = \sum_{m \geq 0}^H \binom{-1/2}{m} (N - \mathbf{1})^m$$

exists by corollary 2.6, is real diagonal, and satisfies $HNH = \mathbf{1}$. Its diagonal entries have leading coefficient 1, hence are positive. Define $U = VH$. Since H commutes with Λ , $AU = UA$ and $U^*U = HNH = \mathbf{1}$. Both V and H are invertible in $\mathcal{A}_I$, so U is invertible and $U^* = U^{-1}$; this proves the other unitary identity as well. Notice that in an infinite algebra a left inverse alone would not have sufficed.

Because $V_{ii} = 1$, one has $U_{ii} = H_{ii} > 0$. The support claims follow from lemma 2.1(ii) applied to the homogeneous expansions; no exponent division is involved in the binomial normalization.

For uniqueness, let (U', Λ') satisfy the same requirements. The intertwining matrix $T = U^*U'$ has residue $\mathbf{1}$. As in the uniqueness proof of theorem 18.1, first $\Lambda' = \Lambda$ and then T is diagonal. It is unitary. From $U'_{ii} = U_{ii}T_{ii}$ and positivity of both diagonal entries, T_{ii} is positive real. The identity $\overline{T_{ii}}T_{ii} = 1$ then forces $T_{ii} = 1$. $\square$

Corollary 18.3 (No new scales). *Assume the hypotheses of theorem 18.2: in particular that $D = \text{diag}(d_i)$ has pairwise distinct ordinary real entries, so that every divisor $d_i - d_j$ used in (18.1) is a nonzero ordinary complex constant of valuation zero, and that $B = B^*$ has globally well-ordered positive support S . Then, under those hypotheses, all eigenvalues λ_i and all entries of the normalized diagonalizer U lie in the original Hahn field K , and their correction supports lie in the additive monoid S^* generated by S ; consequently the result holds over an arbitrary ordered exponent group, with no passage to its divisible hull. If in addition the coefficients of A are real Hahn, the diagonalizer can be taken, and by uniqueness is, real Hahn orthogonal.*

Proof. The support assertion was proved in theorems 18.1 and 18.2. When all coefficients are real, the recursion and the binomial series use real operations throughout. $\square$

Remark 18.4 (The monoid statement is not true without the distinct-residue hypothesis). The support conclusion of corollary 18.3 must not be lifted out of its hypothesis. The collection's finite-dimensional spectral report contains an example, its `spec:ex:monoid`, which shows that the corresponding unconditional statement is false: with $\Gamma = \mathbb{Z}$ and

$$A = \begin{pmatrix} t^3 & t^4 \\ t^4 & 0 \end{pmatrix}, \quad \lambda_+ = t^3 + t^5 - t^7 + 2t^9 - \dots,$$

the exponent 5 is not in the nonnegative monoid generated by the raw input exponents 3 and 4, and that report concludes that a theorem asserting that eigenvalue supports stay in the raw input monoid “would therefore be false” [2].

There is no conflict. Writing that matrix as $D + B$ with B of positive support forces $D = \text{diag}(0, 0)$: both diagonal entries, t^3 and 0 , have residue zero. The residues are therefore not pairwise distinct, the divided commutator $\mathcal{R}_D$ of (18.1) does not exist, and theorems 18.1 and 18.2 and corollary 18.3 do not apply. Corollary 18.3 divides only by differences $d_i - d_j$ of pairwise distinct ordinary complex constants, each of valuation zero; that is precisely what keeps every correction exponent inside S^* . The report cited above states the correct unconditional invariant for the general finite case — the input *subgroup*, supplemented by the positive monoids of the normalized perturbations at each node — and that is a different statement from the one proved here.

18.4 Initial coefficients and their meaning

For a one-scale perturbation $B = t^\eta B_1 + t^{2\eta} B_2 + \dots$, $\eta > 0$, the first terms are

$$\lambda_i = d_i + t^\eta (B_1)_{ii} + t^{2\eta} \left((B_2)_{ii} + \sum_{j \neq i} \frac{|(B_1)_{ji}|^2}{d_i - d_j} \right) + \mathcal{O}(t^{3\eta}), \quad (18.8)$$

$$U_{ji} = t^\eta \frac{(B_1)_{ji}}{d_i - d_j} + \mathcal{O}(t^{2\eta}) \quad (j \neq i), \quad (18.9)$$

$$U_{ii} = 1 - \frac{t^{2\eta}}{2} \sum_{j \neq i} \frac{|(B_1)_{ji}|^2}{(d_i - d_j)^2} + \mathcal{O}(t^{3\eta}). \quad (18.10)$$

Here the remainder notation is a valuation statement within the one-scale series under discussion. Every displayed sum is finite for a fixed i . Equations (18.8)–(18.10) follow directly from (18.3)–(18.7); they are the familiar perturbation coefficients, not independent novelty claims.

What changes in the present theorem is that all orders and all coordinates fit into one operator with globally controlled Hahn support. A uniform bound for the real numbers in (18.9) is unnecessary, but a finite coefficient support in every row and column remains essential to the chosen algebra.

19 Spectral synthesis, commutants, and projections

Throughout this section the hypotheses of theorem 18.2 hold, and $u_i = Ue_i$ denotes its normalized eigenvector.

Theorem 19.1 (Exact Hahn spectral synthesis). *The family $(u_i)_{i \in I}$ is orthonormal. Every $x \in \mathcal{H}_I$ has the unique strongly summable expansion*

$$x = \sum_{i \in I}^H u_i \langle u_i, x \rangle. \quad (19.1)$$

*More precisely, the coordinate map $x \mapsto (\langle u_i, x \rangle)_{i \in I}$ identifies $\mathcal{H}_I$ with itself when the right side is interpreted as the Hahn vector U^*x . One has the identities*

$$\langle x, x \rangle = \sum_{i \in I}^H |\langle u_i, x \rangle|^2, \quad (19.2)$$

$$Ax = \sum_{i \in I}^H \lambda_i u_i \langle u_i, x \rangle. \quad (19.3)$$

The eigenvalues of A on $\mathcal{H}_I$ are exactly the λ_i , and each corresponding eigenspace is Ku_i .

Proof. Orthonormality is the entrywise meaning of $U^*U = \mathbf{1}$. Put $y = U^*x$. At each exponent, y has finite coordinate support. Therefore the family $(e_i y_i)_{i \in I}$ is strongly summable in $\mathcal{H}_I$ and sums to y . Lemma 17.5 gives

$$x = Uy = \sum_i^H Ue_i y_i = \sum_i^H u_i \langle u_i, x \rangle,$$

proving existence and summability. Taking inner products with each u_i proves uniqueness whenever such a strong expansion is given. The diagonal action of Λ takes $\mathcal{H}_I$ to itself because $\Lambda \in \mathcal{A}_I$. Applying U to the coordinate expansion of Λy proves (19.3). The norm identity follows from $\langle x, x \rangle = \langle y, y \rangle$ and proposition 17.3.

Finally, if $Ax = zx$ with $x \neq 0$, then $(\lambda_i - z)y_i = 0$ for all i . At least one y_i is nonzero, so $z = \lambda_i$ for that index. Distinct residues make the λ_i pairwise distinct; all other coordinates of y vanish. This proves the eigenspace statement. $\square$

The expansion need not be finite over K , so (u_i) is generally not a Hamel basis. Nor does (19.2) involve the ordinary convergence of a series of nonnegative real numbers: its terms and sum lie in F , and the assertion is strong Hahn summability.

19.1 The coherent diagonal algebra

Let

$$\mathcal{D}_I = (\mathbb{C}^I)((t^\Gamma)) \subseteq \mathcal{A}_I \quad (19.4)$$

be the algebra of diagonal Hahn operators. Thus a family $(a_i)_{i \in I}$ of elements of K defines an element of $\mathcal{D}_I$ precisely when $\bigcup_i \text{supp } a_i$ is well ordered. There is no finite-incidence condition on the indices i here: an arbitrary diagonal matrix of complex numbers is one legitimate coefficient in R_I .

This is strictly more restrictive than allowing all of K^I , and less restrictive than demanding bounded ordinary diagonal coefficients. Both facts matter for the spectrum calculation.

Theorem 19.2 (The complete commutant). *Within the specified Hahn operator algebra,*

$$\text{Cent}_{\mathcal{A}_I}(A) = U\mathcal{D}_IU^*. \quad (19.5)$$

In particular this commutant is commutative. Its idempotents are exactly

$$P_J = UE_JU^*, \quad J \subseteq I, \quad (19.6)$$

where $E_J = \text{diag}(\mathbf{1}_{i \in J})$. Every such idempotent is self-adjoint. The correspondence $J \mapsto P_J$ is a complete Boolean algebra isomorphism, with complement $\mathbf{1} - P_J$ and product $P_J P_L = P_{J \cap L}$.

Proof. If X commutes with A , then $Y = U^* X U$ commutes with Λ . At entry (i, j) this says $(\lambda_i - \lambda_j)Y_{ij} = 0$. For $i \neq j$ the first factor is nonzero in the field K , so Y is diagonal. Its membership in $\mathcal{A}_I$ is precisely the support coherence in (19.4). The converse is immediate, proving (19.5).

An idempotent of $\mathcal{D}_I$ has entries satisfying $a_i^2 = a_i$ in a field, so each is 0 or 1. Every choice is allowed, since its complete operator support is contained in $\{0\}$. These are exactly the E_J . Conjugating proves the projection statement and the finite Boolean identities. Arbitrary unions and intersections of subsets give the joins and meets for the partial order on idempotents defined by $P \leq Q$ when $PQ = P$; no assertion about an infinite operator-valued series is needed for completeness of this Boolean algebra. $\square$

Corollary 19.3 (Projection sums are pointwise Hahn sums). *For every subset $J \subseteq I$ and every $x \in \mathcal{H}_I$,*

$$P_J x = \sum_{i \in J}^H u_i \langle u_i, x \rangle.$$

If $(J_a)_{a \in L}$ are pairwise disjoint, then $(P_{J_a} x)_{a \in L}$ is strongly summable in $\mathcal{H}_I$ and its sum is $P_{\bigcup_a J_a} x$.

Proof. In the coordinates $y = U^*x$, these are the sums of disjoint coordinate pieces of a Hahn vector. At any exponent, only finitely many of the pieces can be nonzero. Apply lemma 17.5 to transport the sums by U . $\square$

Remark 19.4 (A necessary distinction, and why this is not a spectral measure). For infinite I , the operator family $(P_{\{i\}})_{i \in I}$ is *not* strongly Hahn summable as a family in $\mathcal{A}_I$: its coefficient at exponent zero is $E_{\{i\}} \neq 0$ for every i . Thus

$$\mathbf{1} = \sum_i^H P_{\{i\}}$$

is false under the operator-valued summability definition of section 2.1. The correct assertion is $\mathbf{1}x = \sum_i^H P_{\{i\}}x$ for each $x \in \mathcal{H}_I$. The distinction persists even for $B = 0$ and is not a technical artifact of the proof. It is also the reason theorem 19.2 is not offered as a spectral measure, one of the objects the collection's finite-dimensional report listed as absent and which remains absent here.

19.2 An unrestricted ordinary label calculus

Every function $f : I \rightarrow \mathbb{C}$ defines the operator

$$T_f = U \operatorname{diag}(f(i))U^* \in \operatorname{Cent}_{\mathcal{A}_I}(A).$$

This is a unital involution-preserving homomorphism from $\mathbb{C}^I$ into $\mathcal{A}_I$. No continuity or ordinary boundedness of f is required. It is a *label multiplier calculus*: it does not assign values to an ordinary function at arbitrary surreal arguments. More generally, a K -valued label family is allowed exactly when it belongs to the coherent diagonal algebra $\mathcal{D}_I$. Polynomial expressions $p(A)$ have labels $p(\lambda_i)$ because finite polynomial operations preserve coherence.

20 The exact algebra-relative spectrum and all resolvents

Definition 20.1. For $X \in \mathcal{A}_I$, define

$$\sigma_{\mathcal{A}_I}(X) = \{z \in K : X - z\mathbf{1} \text{ has no two-sided inverse in } \mathcal{A}_I\}.$$

This is a spectrum relative to a unital algebra, with no norm or compactness assumption. It is the second of the two senses recorded in table 1.

Diagonalization alone does not automatically identify this spectrum. Entrywise invertibility of infinitely many diagonal entries does not, in general, guarantee that their inverses have globally well-ordered support. The simple-residue hypothesis gives the additional coherence.

Theorem 20.2 (Coherent resolvents at every noneigenvalue). *Under the hypotheses of theorem 18.2,*

$$\sigma_{\mathcal{A}_I}(A) = \{\lambda_i : i \in I\}. \quad (20.1)$$

For $z \in K$ outside that set,

$$(A - z\mathbf{1})^{-1} = U \operatorname{diag}((\lambda_i - z)^{-1})U^* \in \mathcal{A}_I. \quad (20.2)$$

The conclusion and the proof also hold for the non-Hermitian simple-complex-residue setting of theorem 18.1, with U, U^ replaced by V, V^{-1} .*

Proof. The eigenvector u_i prevents $A - \lambda_i \mathbf{1}$ from being invertible. It remains to prove coherence of the diagonal inverse when every $\lambda_i - z$ is nonzero. Write $\Lambda = D + \Delta$ with $\text{supp } \Delta \subset \Gamma_{>0}$.

First suppose $v(z) < 0$. Then $\Lambda - z\mathbf{1} = -z(\mathbf{1} - z^{-1}\Lambda)$. Since $v(\Lambda) \geq 0$, the operator $z^{-1}\Lambda$ has positive support. The inverse therefore belongs to $\mathcal{D}_I$ by lemma 2.3.

Now suppose $v(z) \geq 0$, also allowing $z = 0$. Write $z = c + \zeta$, where $c \in \mathbb{C}$ is the coefficient at exponent zero and ζ has positive support. If $c \neq d_i$ for every i , the ordinary diagonal matrix $C = \text{diag}((d_i - c)^{-1})$ belongs to R_I . It need not have bounded complex entries, and none are required. The factorization

$$\Lambda - z\mathbf{1} = (D - c\mathbf{1})(\mathbf{1} + C(\Delta - \zeta\mathbf{1}))$$

has a positive-support second factor and an invertible constant first factor. Its inverse is again in $\mathcal{D}_I$.

Finally, if $c = d_j$, then there is exactly one such index j . Split the diagonal algebra into its $E_{\{j\}}$ and $\mathbf{1} - E_{\{j\}}$ parts. On the second part the preceding argument applies, using arbitrary harmless values at the omitted index to write the factorization. On the first part, the inverse is the single scalar Hahn series $(\lambda_j - z)^{-1}$ placed at index j . It exists by hypothesis. The union of the supports of these two pieces is a finite union of well-ordered sets. Adding them gives the desired diagonal inverse. Conjugation by U proves (20.2).

Nothing in the coherence argument used reality of the distinct constants d_i ; the last assertion follows in the same way. $\square$

Example 20.3 (An accumulation point need not be in this spectrum). Take $I = \mathbb{N}_{>0}$, $D = \text{diag}(1/n)$, and $B = 0$. Then 0 is an ordinary accumulation point of the labels, but

$$D^{-1} = \text{diag}(n) \in R_I \subset \mathcal{A}_I.$$

Consequently $0 \notin \sigma_{\mathcal{A}_I}(D)$. This does not contradict the ordinary bounded-operator spectrum on ℓ^2 , where the displayed inverse is unbounded. It records the different algebra in which invertibility is being tested. It also does not contradict corollary 9.2, which puts the whole infinitesimal monad $\mathbf{m}_K$ into the failure-of-bijectivity spectrum of the same diagonal data *as a compact operator on* $\ell^2((t^\Gamma))$; section 3.2 sets the two side by side.

Corollary 20.4 (The inhomogeneous equation). *If $z \notin \{\lambda_i : i \in I\}$, every equation $(A - z\mathbf{1})x = f$ with $f \in \mathcal{H}_I$ has the unique solution*

$$x = \sum_i^H u_i \frac{\langle u_i, f \rangle}{\lambda_i - z}.$$

The series is strongly summable in $\mathcal{H}_I$.

Proof. The coherent inverse diagonal operator takes U^*f to a Hahn vector. Apply U and theorem 19.1. Invertibility gives uniqueness. $\square$

21 Weighted walks control every coefficient

21.1 The graph attached to a Hahn perturbation

Let $S = \text{supp } B \subset \Gamma_{>0}$ and write $B = \sum_{s \in S} B_s t^s$. Construct an edge-labeled directed graph on I by putting an edge

$$j \xrightarrow{s} i \quad \text{when} \quad (B_s)_{ij} \neq 0.$$

Loops are allowed. Since $B_s = B_s^*$, every nonloop edge has a reverse edge with the same label. For a fixed s , every vertex has finitely many incoming and outgoing edges. The graph obtained by forgetting all labels need not be locally finite when S is infinite.

A walk has a finite sequence of edges; its weight is the sum of the positive labels. The empty walk has weight zero. Although an edge label can be infinitesimal relative to another in Γ , no edge has zero or negative weight.

Lemma 21.1 (Finite walks at a prescribed weight). *For fixed starting vertex i and fixed $\gamma \in \Gamma$, there are only finitely many walks beginning at i and having total weight γ . In particular their visited vertices form a finite set.*

Proof. Lemma 2.1(ii) gives finitely many ordered label words with total weight γ . For any one such word, there are finitely many choices of the first edge and, recursively, finitely many choices at each subsequent step. The word length is finite, so the complete set of walks is finite. Taking a finite union over words proves the assertion. $\square$

Theorem 21.2 (Coefficientwise walk dependence). *For the intermediate-normalized V and the normalized unitary U , the coefficient at a positive exponent γ in V_{ji} or U_{ji} is a finite sum of expressions supported on walks from i to j of weight γ . The coefficient of $\lambda_i - d_i$ is a finite sum of expressions supported on closed walks at i of weight γ .*

Each expression is a product of edge coefficients and their conjugates, multiplied by a rational number and by inverse powers of nonzero real label differences at visited vertices. Thus those coefficients are completely determined by the finitely many relevant walks and the ordinary labels at their vertices.

This is a dependence statement, not a claim that every permitted walk contributes a nonzero term, or that its contribution is just its raw edge product.

Proof. Work first with the homogeneous recursion (18.3)–(18.5). A coefficient of $V_0 = \mathbf{1}$ is represented by an empty walk. Multiplication by B attaches an edge to the end of a walk, with matrix multiplication summing over the intermediate vertex. The projection π retains closed walks. In the term $(V_{n-k}\Delta_k)_{ji}$, the scalar $(\Delta_k)_{ii}$ is represented by a closed walk at i ; traverse that closed walk and then a walk from i to j representing $(V_{n-k})_{ji}$. Division by $d_j - d_i$ changes no walk or exponent. This proves the assertion for V and Λ by induction on the homogeneous degree.

For normalization, (18.7) expresses $N_{ii} - 1$ through products $\overline{W_{ki}}W_{ki}$. More generally, at a fixed coefficient one factor can come from one walk from i to k and the other from another. Reverse the conjugated walk, using Hermitian symmetry of the labeled edges. Their concatenation is a closed walk at i . Powers of $N - \mathbf{1}$ concatenate such closed walks; the binomial coefficients contribute only rational scalars. Finally $U_{ji} = V_{ji}H_{ii}$ concatenates those closed walks with a walk from i to j .

Lemma 2.1(ii) bounds the homogeneous degrees at any prescribed exponent, and lemma 21.1 bounds every index sum within them. All coefficient formulas are therefore genuinely finite. $\square$

21.2 Marked-edge stability

Suppose B and $\tilde{B}$ satisfy the same hypotheses with the same D . Their support union S is well ordered and positive. Use the union of the two edge graphs, and mark an edge of label s whenever $(\tilde{B}_s - B_s)_{ji} \neq 0$. For a vertex i , let κ_i be the minimum weight of a closed walk at i containing a marked edge, or $+\infty$ if no such walk exists. The minimum exists whenever the set is nonempty, since its weights form a subset of the well-ordered monoid S^* .

Theorem 21.3 (Exact marked-walk stability). *The eigenvalues labeled by i satisfy*

$$v(\tilde{\lambda}_i - \lambda_i) \geq \kappa_i. \quad (21.1)$$

More generally, the coefficient of $\tilde{U}_{ji} - U_{ji}$ at exponent γ vanishes unless there is a walk from i to j of weight γ containing a marked edge.

In particular, set $E = \tilde{B} - B$. Then

$$v(\tilde{U} - U) \geq v(E), \quad v(\tilde{\Lambda} - \Lambda) \geq v(E).$$

If E has zero diagonal and $S \neq \emptyset$, the second estimate sharpens to

$$v(\tilde{\Lambda} - \Lambda) \geq v(E) + \min S. \quad (21.2)$$

Zero differences are interpreted as having valuation $+\infty$.

Proof. The recursions and the normalization use the same rational expressions in the edge coefficients for both inputs, with identical denominators because D is fixed. Subtract a product of coefficients from its counterpart by successively replacing its factors. Each resulting term contains a factor from E or its conjugate. It therefore has a marked edge in its associated walk. This argument is finite at every exponent by theorem 21.2, so it proves both dependence assertions and (21.1).

A term containing a factor from E has weight at least $v(E)$, since all its other edges have positive weights. If E has zero diagonal, a closed walk containing a marked edge cannot have length one. It has at least one further edge with weight at least $\min S$. This proves (21.2). The same bounds hold uniformly for every matrix coefficient and hence for the global operator order function. $\square$

These statements are valuation precision bounds, not analytic continuity in a real operator norm. They are also stronger than merely saying that finite jets depend on finite jets: they identify *which spatial changes* can first become visible at a given Hahn scale.

21.3 Finite sections with exact coefficient certificates

Let $F_0 \subset I$ be finite and contain i , and let A_{F_0} denote the principal compression of A to F_0 . Its diagonalizer and eigenvalues are normalized by the same rules. Write $\lambda_i^{F_0}$ for its eigenvalue with residue d_i . Define the escape weight

$$\kappa_i(F_0) = \min\{\text{wt}(w) : w \text{ is a closed walk at } i \text{ and visits } I \setminus F_0\},$$

again using $+\infty$ for an empty set.

Theorem 21.4 (Finite-section exactness). *For every finite $F_0 \ni i$,*

$$v(\lambda_i - \lambda_i^{F_0}) \geq \kappa_i(F_0). \quad (21.3)$$

For a fixed exponent γ , the coefficient of λ_i is exactly that of a finite principal section containing all vertices of closed walks at i of weight γ . The analogous statement for U_{ji} holds with walks from i to j and with both endpoints included. Consequently every single requested coefficient has a finite spatial certificate, even when I and $\text{supp } B$ are infinite.

Proof. For the coefficient assertion, all relevant walks are finite in number by lemma 21.1. If their vertices lie in F_0 , every finite expression in theorem 21.2 is unchanged by compression. In particular, every coefficient strictly below $\kappa_i(F_0)$ agrees, which proves (21.3).

Alternatively, remove all edges crossing between F_0 and its complement while keeping the diagonal and all internal edges. The resulting infinite operator is a direct sum on those two index sets. Its recursion restricted to F_0 is exactly the recursion for A_{F_0} . A closed walk at i contains a removed crossing edge exactly when it leaves F_0 . Thus theorem 21.3 gives the same inequality. $\square$

The finite section for one coefficient need not work for *all* coefficients below a prescribed arbitrary-rank exponent. Such an initial segment can be infinite and can involve infinitely many spatial vertices. The theorem guarantees finiteness at each exact weight; it does not replace an infinite Hahn support by a finite truncation without additional data.

Corollary 21.5 (One-scale radius bound). *Let $A = D + t^\eta C$, where $\eta > 0$ and $C = C^* \in R_I$, and let F_0 contain the graph ball of radius r about i in the undirected support graph of C . Then*

$$v(\lambda_i - \lambda_i^{F_0}) \geq (2r + 2)\eta. \quad (21.4)$$

The coefficient of $t^{n\eta}$ in λ_i depends only on the ball of radius $\lfloor n/2 \rfloor$ about i . The coefficient of $t^{n\eta}$ in U_{ji} vanishes when $\text{dist}(i, j) > n$.

Proof. A walk that leaves the radius- r ball requires at least $r + 1$ edges to reach the outside and at least $r + 1$ more to return. Every edge has weight η . A closed walk of length n cannot visit a vertex farther than $\lfloor n/2 \rfloor$ from its starting point, while a walk from i to j has length at least their graph distance. Apply theorems 21.2 and 21.4. $\square$

21.4 The threshold is sharp

Theorem 21.6 (A nonzero first boundary coefficient). *Let $A_L(t)$ be the tridiagonal matrix on vertices $0, \dots, L$ with pairwise distinct real diagonal entries $d_0, \dots, d_L$ and off-diagonal entries tb_k and $\overline{tb_k}$ between k and $k + 1$, where $b_k \in \mathbb{C} \setminus \{0\}$. Let $\lambda_0^{[L]}$ be its branch with residue d_0 , and compare it with the corresponding branch for the section $0, \dots, L - 1$. Then*

$$\lambda_0^{[L]} - \lambda_0^{[L-1]} = \frac{\prod_{k=0}^{L-1} |b_k|^2}{(d_0 - d_L) \prod_{k=1}^{L-1} (d_0 - d_k)^2} t^{2L} + \mathcal{O}(t^{2L+2}). \quad (21.5)$$

In particular the leading coefficient is nonzero, and the exponent in (21.4) cannot be increased in general.

Proof. Use $q = t^2$ and put $c_k = |b_k|^2$. Schur elimination on the tail gives the scalar equation

$$x - d_0 = qc_0 H_1^{[L]}(x, q), \quad H_k^{[L]}(x, q) = \frac{1}{x - d_k - qc_k H_{k+1}^{[L]}(x, q)}, \quad (21.6)$$

with $H_L^{[L]} = (x - d_L)^{-1}$. For x near d_0 , all constant terms of the denominators are nonzero, so these are formal power series in q with coefficients rational in x , regular at d_0 .

For $L \geq 2$, subtract the full and truncated tail fractions. At the last common vertex one obtains

$$H_{L-1}^{[L]} - H_{L-1}^{[L-1]} = \frac{qc_{L-1}}{(x - d_{L-1})^2(x - d_L)} + \mathcal{O}(q^2).$$

At each preceding vertex, the identity

$$\frac{1}{a - qcH} - \frac{1}{a - qc\tilde{H}} = \frac{qc(H - \tilde{H})}{(a - qcH)(a - qc\tilde{H})}$$

adds one power of q , one factor c , and a squared leading denominator. Thus

$$qc_0(H_1^{[L]} - H_1^{[L-1]}) = \frac{\prod_{k=0}^{L-1} c_k}{(x - d_L) \prod_{k=1}^{L-1} (x - d_k)^2} q^L + \mathcal{O}(q^{L+1}).$$

The two root equations have derivative $1 + \mathcal{O}(q)$ with respect to x at their solutions near d_0 . Subtracting the equations therefore gives the same first root difference, evaluated at $x = d_0$. For $L = 1$ the truncated branch is exactly d_0 , and the original Schur equation gives the same formula directly. This proves (21.5). The nonzero denominator and the nonzero positive numerator prove sharpness. $\square$

22 An accumulating-level chain with an exact eigenpair

Consider $I = \mathbb{N}_0$, $\Gamma = \mathbb{Z}$, and the real symmetric operator

$$(Ax)_n = \frac{x_n}{n+1} + t(x_{n-1} + x_{n+1}), \quad x_{-1} = 0. \quad (22.1)$$

Its diagonal labels $d_n = 1/(n+1)$ are pairwise distinct but accumulate at zero, and the neighboring gaps

$$d_n - d_{n+1} = \frac{1}{(n+1)(n+2)}$$

have no uniform positive real lower bound. Theorem 18.2 nevertheless constructs all its normalized eigenvectors simultaneously in $\mathbb{C}^{(\mathbb{N}_0)}((t))$.

22.1 The first eigenvalue can be written in closed form

Proposition 22.1 (An exact branch and its eigenvector). *For the operator (22.1), the eigenvalue with residue $d_0 = 1$ is*

$$\lambda_0 = \sqrt{1 + 4t^2}, \quad (22.2)$$

where the square root has residue 1. Define

$$q = \frac{\lambda_0 - 1}{2t} = \frac{2t}{\lambda_0 + 1}. \quad (22.3)$$

Then $v(q) = 1$, the intermediate-normalized eigenvector has coordinates $(v_0)_n = (n+1)q^n$, and the normalized eigenvector is

$$(u_0)_n = (n+1)q^n \left(\frac{(1-q^2)^3}{1+q^2} \right)^{1/2}. \quad (22.4)$$

All square roots in these formulas are near-one binomial Hahn roots.

Proof. The binomial series defines (22.2); it gives $q = t + \mathcal{O}(t^3)$ and the exact identities

$$t = \frac{q}{1-q^2}, \quad \lambda_0 = \frac{1+q^2}{1-q^2} = t(q+q^{-1}). \quad (22.5)$$

For $n \geq 0$, direct substitution gives

$$\begin{aligned} \frac{(n+1)q^n}{n+1} + t(nq^{n-1} + (n+2)q^{n+1}) &= q^n(1 + t(nq^{-1} + (n+2)q)) \\ &= \lambda_0(n+1)q^n. \end{aligned}$$

For $n = 0$, the term with the factor n is zero, agreeing with the boundary condition. Because $v(q^n) = n$, at any exponent only finitely many vector coordinates can contribute. The displayed sequence is therefore a Hahn vector, not merely an arbitrary sequence of elements of K . It has zeroth coordinate 1 and residue e_0 . The eigenspace statement in theorem 19.1, together with its residue and zeroth coordinate, identifies it with the unique intermediate-normalized branch constructed in theorem 18.1.

The elementary formal identity

$$\sum_{n \geq 0} (n+1)^2 z^n = \frac{1+z}{(1-z)^3}$$

can be evaluated at $z = q^2$ by corollary 2.6. Hence

$$\langle v_0, v_0 \rangle = \frac{1+q^2}{(1-q^2)^3}.$$

Taking its inverse near-one square root gives (22.4). $\square$

Proposition 22.1 is a directly proved explicit example; no separate priority claim is made for this solvable tridiagonal model. It also checks the perturbation construction by a method independent of the matrix recursion. Its eigenvalue begins

$$\lambda_0 = 1 + 2t^2 - 2t^4 + 4t^6 - 10t^8 + 28t^{10} + \mathcal{O}(t^{12}). \quad (22.6)$$

The included exact computation obtains these same coefficients from the six-vertex principal section. Corollary 21.5 shows that the section $0, \dots, 5$ gives the infinite branch correctly through degree 10; its first possible error is degree 12.

22.2 The first finite-section error is explicit

For the path section $0, \dots, L$ with unit edge coefficients, the coefficient in theorem 21.6 simplifies to

$$\frac{1}{(1 - 1/(L+1)) \prod_{k=1}^{L-1} (1 - 1/(k+1))^2} = L(L+1).$$

Indeed, $1 - 1/(k+1) = k/(k+1)$ and the product telescopes. Therefore

$$\lambda_0^{[L]} - \lambda_0^{[L-1]} = L(L+1)t^{2L} + \mathcal{O}(t^{2L+2}). \quad (22.7)$$

The infinite branch and the section $0, \dots, L$ agree through degree $2L$, so the same first coefficient describes the error made by truncating the infinite chain at $L-1$.

This is stronger than a qualitative finite-section limit: it specifies the first scale at which the omitted boundary is visible and gives its nonzero coefficient. It makes sense without any ordinary numerical choice of a small real t .

23 A higher-rank example: exact sums without cofinal powers

Let $\Gamma = \mathbb{Z}^2$ with lexicographic order, and set $a = (0, 1)$, $b = (1, 0)$. Both are positive, but $na < b$ for every positive integer n . Consider

$$A = \begin{pmatrix} 0 & t^a & 0 \\ t^a & 1 & t^b \\ 0 & t^b & 3 \end{pmatrix}. \quad (23.1)$$

The normalized branch at vertex 0 has the coefficient identities

$$[t^{2a}]\lambda_0 = -1, \quad [t^{a+b}]U_{20} = \frac{1}{3}, \quad [t^{2a+2b}]\lambda_0 = -\frac{1}{3}. \quad (23.2)$$

The first uses the edge $0 \leftrightarrow 1$. The distant eigenvector coordinate uses the path $0 \rightarrow 1 \rightarrow 2$, and the last eigenvalue coefficient is the first return through the distant edge $1 \leftrightarrow 2$. These coefficients follow from the recursion, or from the shortest corresponding walk expressions; the included bivariate calculation checks them exactly.

There can be infinitely many powers of t^a before the scale t^b . Accordingly, a finite total-degree computation in two formal variables is *not* the same as computing every coefficient below b in the lexicographic valuation. The verification program checks specified bidegrees and explicitly does not conflate these truncations.

Proposition 23.1 (Why geometric partial sums need not converge). *In this Hahn field, the family $((t^a)^n)_{n \geq 0}$ is strongly summable with sum $(1 - t^a)^{-1}$, but its finite partial sums do not converge to that sum in the valuation topology of K .*

Proof. Lemma 2.3 gives the sum and its algebraic identity. The exact remainder after degree N is

$$\frac{t^{(N+1)a}}{1 - t^a},$$

of valuation $(N+1)a < b$. Thus it never belongs to the neighborhood of zero defined by $v(x) > b$. This prevents convergence to the Hahn sum. $\square$

In a surreal realization sending a to 1 and b to ω , the monomials become ω^{-1} and $\omega^{-\omega}$. The same exact remainder remains larger in magnitude than the latter infinitesimal scale. The spectral theorem uses the word-finiteness proof, so it remains valid despite this failure of sequential convergence.

24 Three boundaries that cannot be omitted

24.1 No bounded Hilbert-space diagonalizer with these jets

Proposition 24.1 (The inverse-gap obstruction). *On ordinary $\ell^2(\mathbb{N}_{>0})$, let $D = \text{diag}(1/n)$ and let C be the nearest-neighbor adjacency operator. Both D and C are bounded and self-adjoint. There is no formal diagonalization of $D + tC$ of the form*

$$(D + tC)U(t) = U(t)\Lambda(t), \quad U(t) = \mathbf{1} + tX + \cdots,$$

with X a bounded operator and $\Lambda(t)$ diagonal with constant term D . In particular no ordinary operator-norm analytic near-identity unitary diagonalizer can have that property.

Proof. The order- t off-diagonal equation is $[D, X] + C = \text{diagonal}$. At the neighboring entry $(n, n+1)$ it gives

$$X_{n,n+1} = -\frac{1}{1/n - 1/(n+1)} = -n(n+1).$$

Every matrix entry of a bounded operator has absolute value at most its operator norm. The displayed sequence is unbounded, a contradiction. An operator-norm analytic family would have a bounded derivative at zero, so it is ruled out as well. $\square$

The diagonalizer of theorem 18.2 does have this first coefficient, and that coefficient belongs to R_I : it is supported on neighboring diagonals, even though its real entries are unbounded. Thus the theorem does not secretly solve the ordinary small-divisor or bounded-operator problem. It solves a different, explicitly algebraic problem with exact infinitesimal scales. The same contrast appears from the other side in section 14, where the excluded object is an unbounded diagonal *coefficient* of an adjointable operator.

24.2 Repeated residues can destroy every eigenvector

Let T be the bilateral adjacency operator on finite-support vectors indexed by $\mathbb{Z}$:

$$(Tx)_n = x_{n-1} + x_{n+1}.$$

It is real symmetric and row- and column-finite. A nonzero finite-support vector cannot be an eigenvector of T : at the index immediately above the largest nonzero coordinate, Tx is nonzero while the vector itself is zero. The same observation shows that T is injective on $\mathbb{C}^{(\mathbb{Z})}$.

Proposition 24.2 (A positive Hahn operator with no eigenvectors). *For any $\eta \in \Gamma_{>0}$, the operator*

$$A = \mathbf{1} + t^\eta T \in \text{RCF}_{\mathbb{Z}}(\mathbb{C})((t^\Gamma))$$

is self-adjoint and satisfies $\langle x, Ax \rangle > 0$ for every nonzero $x \in \mathbb{C}^{(\mathbb{Z})}((t^\Gamma))$. Nevertheless it has no eigenvalue in K for its action on that Hahn vector space.

Proof. If $\gamma = v(x)$, the leading term of $\langle x, x \rangle$ has exponent 2γ and positive coefficient. The term $t^\eta \langle x, Tx \rangle$ has valuation at least $2\gamma + \eta$. Thus $\langle x, Ax \rangle$ has the same positive leading term as the norm square.

Suppose $Ax = zx$ with $x \neq 0$. Injectivity of T on the leading finite-support coefficient gives $v(Tx) = v(x) = \gamma$. Hence $z \neq 1$ and $v(z - 1) = \eta$ in the equation $t^\eta Tx = (z - 1)x$. If c is the nonzero leading coefficient of $t^{-\eta}(z - 1)$, comparison at exponent $\gamma + \eta$ gives $Tx_\gamma = cx_\gamma$. That is a forbidden finite-support eigenvector of the bilateral adjacency operator. The contradiction proves the assertion. $\square$

All residue labels in this example are 1. Thus replacing “pairwise distinct ordinary residues” in theorem 18.2 merely by “positive self-adjoint” would be false. The general phenomenon of positive infinite row- and column-finite matrices without eigenvalues is already emphasized in [16]; the proof here shows explicitly how it survives the chosen Hahn extension.

24.3 Entrywise Hahn matrices are not enough

Proposition 24.3 (Failure without global support coherence). *Let $I = \mathbb{N}_0$ and $\Gamma = \mathbb{Z}$. The entrywise Hahn diagonal matrix $M = \text{diag}(t^{-n})_{n \geq 0}$ does not define an operator from $\mathcal{H}_I$ to itself by coordinatewise multiplication, although every entry belongs to K and every individual exponent coefficient is row- and column-finite.*

Proof. Its union of supports is $\{0, -1, -2, \dots\}$, which is not well ordered; thus $M \notin \mathcal{A}_I$. More concretely,

$$x = \sum_{n \geq 0} t^n e_n \in \mathcal{H}_I$$

would be sent to the sequence $(1, 1, 1, \dots)$. This would require an infinite-support coefficient at exponent zero, so it is not an element of $\mathcal{H}_I$. $\square$

Likewise, a coefficient matrix with an infinite nonzero column can send one e_i outside $\mathbb{C}^{(I)}$; requiring row-finiteness as well makes the adjoint remain in the same algebra. These hypotheses are simple sufficient conditions for the whole framework. The counterexamples do not claim that no larger useful operator categories exist.

25 Transport to surreal numbers and extension of scales

Corollary 25.1 (An infinite surcomplex spectral system). *Whenever the input is given in a workspace embedded by (2.2), every scalar entry of U , every eigenvalue λ_i , and every scalar coordinate of a vector in $\mathcal{H}_I$ is an actual surreal or surcomplex number. All identities in sections 17 to 21 remain valid after this embedding, with sums interpreted as the transported strong sums. The eigenvalues of a Hermitian input are surreal, and its inner products have positive surreal norm squares.*

Proof. The embedding preserves every finite coefficient operation and every strong sum used in the construction. It preserves order and conjugation, so it preserves positivity, adjoints, and the normalization conditions. $\square$

The matrices and vectors remain set-indexed structured objects; an infinite matrix is not being identified with one surcomplex scalar. Nor is this a theorem about every conceivable class-sized operator over $\mathbf{No}[i]$. It provides a rigorous infinite-dimensional theory in a fixed, explicitly embedded workspace.

Proposition 25.2 (Naturality and absence of new spectral points). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be a strictly increasing additive embedding. Apply it to all Hahn exponents of A , keeping the same index set and complex coefficients. The normalized diagonalizer and eigenvalues in the Γ' -workspace are exactly the images of U and λ_i . Moreover,*

$$\sigma_{A_{I,\Gamma'}}(\iota_* A) = \{\iota_* \lambda_i : i \in I\}.$$

In particular, enlargement of the value group creates neither new labeled eigenvalues nor additional algebraic spectral points for this simple-residue class.

Proof. An order embedding takes well-ordered supports to well-ordered supports and preserves the finiteness of coefficient incidences. Hence it commutes with all strong sums, products, divided commutators, and near-one binomial operations in the proof of theorem 18.2. The transported pair satisfies the same normalization, and uniqueness identifies it with the pair constructed in the larger workspace. Theorem 20.2 applied there, including to parameters $z \in K_{\Gamma'}$ outside the smaller field, proves the displayed equality. $\square$

This is an algebraic invariance statement. A scale extension can change whether a sequence of partial sums converges in the intrinsic valuation topology, as proposition 23.1 illustrates. The exact Hahn identities are stable even when such sequential behavior changes. Corollary 9.3 is the corresponding statement in Part I; it concerns a different construction, is restricted to divisible groups, and is proved separately.

26 Effective coefficient calculation

26.1 What a certified computation needs

The proof gives an exact coefficient algorithm once the input presentation supplies the required finite data. To compute a coefficient at a weight γ , one needs a finite list of all ordered input-weight words summing to γ , the coefficient matrices on the finitely many vertices those words

can visit, and exact arithmetic for the ordinary labels and edge coefficients. Theorem 21.2 then reduces the problem to finite rational operations and finite matrix jets.

For a one-scale locally finite graph and an integer degree n , the spatial part is particularly explicit: use the radius- $\lfloor n/2 \rfloor$ ball for an eigenvalue coefficient, and a sufficiently large radius- n ball for an eigenvector coefficient. Equations (18.3)–(18.5) compute the intermediate normalization, after which (18.7) computes the unitary normalization.

For an arbitrary abstract ordered group, the existence of finitely many weight decompositions does *not* provide an algorithm that finds them from an order-comparison oracle. Likewise, arbitrary complex coefficients need not have a decidable exact equality test. The theorem is therefore an exact mathematical construction with a stated computation contract, not a universal effective representation of all surreal numbers.

26.2 A coefficient-level description of the implementation

The Part II program uses exact SymPy matrices with rational and Gaussian rational entries. It implements finite multivariate Taylor jets, indexed by tuples $\alpha \in \mathbb{N}^r$ of bounded total degree. With $B = \sum_{\beta \neq 0} B_\beta z^\beta$, it computes

$$T_\alpha = \sum_{0 < \beta \leq \alpha} B_\beta V_{\alpha-\beta} - \sum_{0 < \beta < \alpha} V_{\alpha-\beta} \Delta_\beta,$$

$$\Delta_\alpha = \pi(T_\alpha), \quad V_\alpha = -\mathcal{R}_D(T_\alpha).$$

The second sum means componentwise $\beta \leq \alpha$ with both β and $\alpha - \beta$ nonzero. It next forms $N = V^*V$ and evaluates the truncated binomial series for $N^{-1/2}$. All coefficients retained are checked against the full eigenidentity and both unitary identities.

The program does not numerically approximate eigenvalues. It also does not pretend to enumerate arbitrary Hahn supports or implement infinite matrices. The infinite claims rest on the proofs in this part; the code tests finite algebraic instances and the advertised coefficient formulas.

27 Part II: further questions and a formalization route

27.1 Beyond simple ordinary residue labels

The repeated-residue counterexample proposition 24.2 shows that infinite degeneracy cannot be ignored. A natural next class consists of diagonal residue blocks that are finite, perhaps with unbounded block sizes, together with an explicit hierarchy of positive splitting scales. A useful theorem would need a global support certificate ensuring that the block refinements and their unitary transformations remain coherent across all blocks. Finite-dimensional splitting by itself does not provide that uniform certificate.

Another direction replaces ordinary gaps by Hahn-valued gaps. Then dividing by $d_i - d_j$ can lower valuations by index-dependent amounts. A possible hypothesis would control the total gap losses along weighted walks so that the resulting effective weights are still well ordered and have finite word multiplicity. The present proof does not establish that extension: its divided commutator is support preserving exactly because the nonzero gaps are ordinary constants. Remark 18.4 is the concrete warning attached to this point.

27.2 Larger vector and operator categories

The vector space $\mathcal{H}_I$ admits only finite-support coefficients. Allowing ordinary ℓ^2 coefficients would include many more states, but real small divisors can then invalidate the recurrence,

as proposition 24.1 shows. A weighted Hilbert-coefficient or Banach-coefficient Hahn category would have to replace row/column finiteness by analytic estimates compatible with the inverse commutator. Such an extension could connect the exact algebraic results to ordinary spectral approximation, but it is not a consequence of the current construction. Part I is a Hilbert-coefficient theory in that spirit, but for a different class of operators, and it does not supply the missing diagonalization theorem either.

Similarly, the marked-walk bounds identify first possible errors, not their noncancellation in every graph. It is worthwhile to classify graphs and signs for which the minimum escaping-walk weight always survives in the coefficient difference. The path formula (21.5) proves this only for a particular family.

27.3 A proof-assistant decomposition

A formalization can be organized without initially constructing the entire surreal class. The foundational objects are the set-sized algebra R_I , its involution and finite-support module, followed by their Hahn extensions. The essential reusable ingredients are: finite row/column support under products; the inner-product leading-term lemma; ordered-word summability; coefficientwise divided commutators; the homogeneous diagonalization recursion; and the uniqueness argument.

After those, normalization, synthesis, and commutant classification are short algebraic developments. The resolvent theorem needs a separate coherence lemma for infinitely many inverse diagonal entries, while locality requires an inductive relation between coefficient expressions and finite labeled walks. Only the final transport step needs a proved normal-form embedding into actual surreal numbers. No claim is made that these new infinite-dimensional statements have been added to, compiled in, or checked by the repository.

27.4 Conclusion of Part II

The principal result is not that every infinite Hermitian surcomplex matrix has a spectral theorem. That assertion is false in the category defined here, by proposition 24.2. The result is that simple ordinary residues and global positive Hahn support provide a complete, exact, infinite-dimensional spectral mechanism with unusually strong locality and extension properties. Its infinitesimal series are actual algebraic objects, its vector expansions are licensed sums, its resolvents are coherent operators, and its finite-section errors have verifiable scale thresholds.

Shared back matter

28 The two exact check suites

The two parts arrived with separate verification programs, which are kept separate in `code/` and `data/`. Neither program checks any statement of the other part.

Warning 28.1 (A coincidence, not a dependence). Both suites report exactly **218 exact checks**, and both recorded runs used **Python 3.13.5** with **SymPy 1.14.0**. This is a coincidence of two independently written programs, and the shared tool versions are a coincidence of two runs on the same day. The two counts are of entirely different assertions: the Part I total is 218 assertions in six groups about finite convolutions, adjoints, truncated Neumann identities, the isolated-point resolvent, a finite rectangular analogue, finite sections and a backward-shift residual, while the Part II total is 218 checks about perturbation jets, an accumulating chain, sharp boundary coefficients, bidegree truncations, projection identities and a closed-form eigenpair. Neither number is evidence about the other suite, and neither program copied the other.

28.1 The Part I suite

The program `code/02-hahn-hilbert-spectral-verify.py` uses exact rational and symbolic arithmetic. It checks finite-support convolution and adjoints, including a lexicographically ordered two-dimensional exponent group; truncated noncommutative Neumann identities; the isolated-point resolvent formula (8.4); the algebraic left-inverse and complement projection identities of theorem 7.1 in a finite rectangular analogue; the finite-section formulas of section 14; and the backward-shift residual of example 8.4. The recorded run passed 218 exact assertions in six groups; its output is `data/02-hahn-hilbert-spectral-verification.txt`.

The finite rectangular analogue is deliberately identified as such. An injective nonsurjective endomorphism cannot exist on a finite-dimensional space, so a finite square-matrix test cannot validate the infinite range defect. Nor can finite tests establish Baire category, arbitrary Hahn support well-ordering, an infinite cokernel dimension, or the spectral formula. Those assertions rest on the proofs, not on the program.

28.2 The Part II suite

The program `code/03-infinite-spectral-verify.py` passed **218 of 218** exact checks; the machine-readable record is `data/03-infinite-spectral-verification.json`. The suite verifies a complex Hermitian three-by-three example with two perturbation orders through degree six; the six-vertex accumulating-level chain through degree ten; the sharp finite-section coefficient for path distances one through four; a two-variable example through total degree four, including the coefficients in (23.2); projection idempotence and self-adjointness at every retained bidegree; the symbolic inverse-gap obstruction of proposition 24.1; and the recurrence, quadratic identity, normalization generating function, and Taylor coefficients of the closed-form eigenpair of proposition 22.1.

These are exact algebraic checks, not floating-point evidence for an infinite theorem. They are not a Lean formalization, do not certify novelty, and cannot replace the summability arguments. The script raises an exception at the first failed check. The two-scale test uses finite bidegrees, not an allegedly finite initial segment of a lexicographic valuation group.

28.3 Building

The L^AT_EX source contains its bibliography and needs no external images, fonts, or bibliography database. A standard build is

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python code/02-hahn-hilbert-spectral-verify.py
python code/03-infinite-spectral-verify.py
```

Both Python scripts need SymPy and support Python 3.10 or later. The source is not Lean code, and no proof-assistant certification is claimed for either part.

A Part I: repository and literature audit

A.1 Repository snapshot

The Part I coverage comparison is pinned to

608dd23c0539fce73723843949ee627641c27b30.

It inspected the repository tree, the project README.md, the full docs/README.md catalogue, the opening 250 lines of docs/surcomplex/spectral-theory/article.tex, and the opening 230 lines of the differential-equations article. This was a targeted scope audit, not a claim to have reread every archived source manuscript or verified every existing Lean file. The spectral opening explicitly states its finite-dimensional scope, while the catalogue maps the already-covered analytic, differential, Berkovich, foundational and computational strands. No repository files were modified by that audit.

A.2 Primary-source comparison and date boundary

The Part I literature search was carried out through September 22, 2026. The publisher records and accessible abstracts for [9, 10], the author-provided text associated with [8], and the arXiv records for [11, 12] were used to locate nearby topics. Ishizuka’s paper appears in volume 39(2), labeled 2024, with the publisher recording online publication on January 3, 2025. Both dates are recorded in the bibliography to avoid silently conflating the issue year with the online date.

The comparison supports the modest statement that the exact theorem package of Part I was not identified in the inspected sources. It does not establish that no earlier work contains any equivalent result. In particular, formal support inversion, failure of non-Archimedean orthogonality, and ordinary resolvent expansions have substantial pre-existing literatures.

B Part II: provenance and research scope

B.1 Repository snapshot

The Part II comparison is pinned to

a3124af79f66b8b9c196d76b4cbc5ac3938907c4

with commit timestamp 2026-09-22T15:58:14Z as returned by GitHub. It used the documentation catalogue and the spectral-theory README at that commit; the differential-equations and

spectral-theory inventories were also examined to avoid repeating finite-dimensional work. The pinned spectral-theory scope explicitly restricts dimensions to ordinary finite integers. This is a targeted scope comparison, not a line-by-line audit of every theorem in the repository. Initially returned unpinned documentation was older; the provenance is based on the pinned snapshot above. No change was made to the GitHub repository.

The two audits used two different pinned revisions, and neither was repeated against the other's pin when the two manuscripts were assembled here.

B.2 Primary sources consulted

Gonshor for foundational normal forms [4]; Neumann [5]; Higman [6]; Greenbaum, Li and Overton [13]; Kenmoe, Smerlak and Zadorin [14]; Giscard, Thwaite and Jaksch [15]; and Ara, Goodearl and O'Meara [16], particularly for the operator-category distinctions and failures of general infinite-dimensional spectral assertions.

The source list and comparison do not certify the absence of all prior versions of the new package. The elementary perturbation coefficients, the formal recursion mechanism, the Neumann support lemma, the Higman word lemma, the walk interpretation of products, and the broad phenomenon of positive infinite matrices with no eigenvectors are not claimed as discoveries.

B.3 Candidate contribution

The Part II candidate-original contribution is the global arbitrary-rank Hahn operator/module synthesis: unique normalized infinite diagonalization with support control, all coherent resolvents, a complete commuting projection calculus, marked-walk stability, and sharp finite-section error thresholds, under simple ordinary residues and positive global support. The explicit accumulating-level closed-form eigenpair of proposition 22.1 is proved directly and used as an independent check; no separate priority claim is made for that solvable model.

B.4 A proof not reprinted

The Part II source manuscript carried an appendix with the classical minimal-bad-sequence proof of Higman's lemma, specialized to a well-ordered alphabet, so that the summability argument would be auditable. That proof is not reprinted here, because the identical argument is already written out in this collection, as `spec:lem:words` and `spec:lem:neumann` of the finite-dimensional spectral report [2]. Lemma 2.1 above states what both parts use, and strict positivity of the letters in an ordered group is an additional hypothesis used only after the word-order result, to pass from embeddings to comparisons of sums.

C Dependency and proof-obligation ledgers

C.1 Part I

All general statements in theorem 1.1 have proofs in the text, relative to the named classical inputs. No computational observation is used as a premise of those proofs. The most consequential steps to audit independently are the Baire uniform-support argument, the complement projection in theorem 7.1, and the accumulation-point case in theorem 8.2; their details are given without an appeal to a numerical experiment.

Result	Inputs	Independent restriction
Positive inner product and norm	Finite Hahn convolution; even leading exponent; binomial root	No divisibility needed
Automatic adjoint structure	Ordinary closed graph; Baire; Hahn support	Everywhere-defined adjoints
Defect splitting	Injective leading term; Neumann support lemma	No topology or commutation needed
Normal spectral formula	Defect splitting; ordinary normal spectral theorem	Constant normal operator
New eigenvalue exclusion	Orthogonal eigenspace projection	Normality is essential
Continuum cokernel	Leading-coefficient obstruction; power-series divergence test	Ordinary ℓ^2 coefficients
Closed coercive range	Defect valuation equality; metric completeness	Rank-one value group
Least norm-bound criterion	Ordinary positive square root; leading norm ratio	Constant bounded operator

Table 3: Where each Part I hypothesis enters; “complete” does not replace “orthogonally complete.”

C.2 Part II

The logical dependence of the Part II claims is as follows. The Hahn algebra and inner product depend on finite coefficient convolution, not on any spectral theorem. Ordered-word finiteness licenses the homogeneous construction. Simple residues make the divided commutator invertible on off-diagonal coefficients. Self-adjointness and positivity make the constructed eigenvalues real and permit near-one unitary normalization. The invertibility already proved for the normalization is what supplies two-sided unitarity. Synthesis uses vector-valued strong sums. Commutant classification uses only the pairwise nonzero eigenvalue differences. The full spectrum theorem adds a separate inverse-coherence argument. Locality uses the homogeneous formulas and finite weighted walks, and surcomplex transport uses the classical normal-form embedding.

The Part II results do not assert a solution to a named published open problem; uniqueness or priority of every underlying idea; existence of a bounded operator-norm diagonalizer; a Hilbert-space or probability interpretation; convergence of the Hahn series in the fine surreal topology or in every intrinsic valuation topology; an operator-valued Hahn sum of all spectral projections; a spectral theorem for arbitrary repeated residues; coherence of arbitrary entrywise Hahn matrices; decidability for arbitrary surreal presentations; or a completed Lean formalization.

C.3 What this report as a whole does not claim

Nothing above unifies the two parts into a single theorem, and no hypothesis is transferred between them. In particular, the divisibility of Γ assumed throughout Part I is not assumed anywhere in Part II, and the row- and column-finiteness of Part II is not available in Part I. The two spectra of table 1 are not proved to agree, and no claim is made that either determines the other. Neither part is a trace-class theory and neither supplies a spectral measure.

References

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